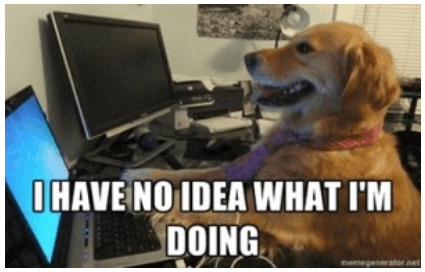
Por fin tenemos reference.



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1 Descripción de Algoritmos

1.1 Data structures

Segment tree — **Complexity:** $O(Nbuild, \log Nperoperation)$

Usage: 1. `STree(int n)` → construye el arbol para n posiciones. 2. `upd(int p, int v)` → actualiza la posicion p con el valor v. 3. `query(int l, int r)` → devuelve el minimo en el rango [l, r). *Note:* Indexacion base 0; adaptar operacion neutra y combine para otro monoid.

Segment tree lazy — **Complexity:** $O(Nbuild, \log Nperupdate/query)$

Usage: 1. `STree(int n)` → crea un arbol con lazy propagation. 2. `upd(int l, int r, int v)` → suma v al rango [l, r). 3. `query(int l, int r)` → devuelve la suma del rango [l, r). *Note:* Ejemplo de suma con incremento por rango; cambiar tipo de lazy segun el problema.

Persistent segment tree — **Complexity:** $O(\log Nperversionupdate/query)$

Usage: 1. `STree(int n)` → reserva el arbol persistente sobre n posiciones. 2. `upd(int root, int p, int v)` → crea una nueva version cambiando la posicion p. 3. `query(int root, int l, int r)` → consulta el minimo en [l, r) para una version. *Note:* Mantener los indices de raiz de cada version por fuera.

2D segment tree — **Complexity:** $O(NMbuild, \log N \log Mperoperation)$

Usage: 1. `build()` → construye el arbol sobre la matriz global. 2. `upd(int x, int y, int v)` → actualiza una celda. 3. `query(int x0, int x1, int y0, int y1)` → consulta el rectangulo [x0, x1) x [y0, y1). *Note:* Usa arreglos globales; ajustar N y M antes de copiar.

Sparse table — **Complexity:** $O(N \log Nbuild, 1query)$

Usage: 1. `st_init(int *a)` → precalcula la tabla para el arreglo a. 2. `st_query(int s, int e)` → devuelve el minimo en [s, e). *Note:* Solo sirve para operaciones idempotentes como min/max/gcd.

Fenwick tree — **Complexity:** $O(\log Nperupdate/query)$

Usage: 1. `upd(int i0, int v)` → suma v a la posicion i0. 2. `get(int i0)` → devuelve la suma de [0, i0). 3. `get_sum(int i0, int i1)` → devuelve la suma de [i0, i1). *Note:* API externa base 0; internamente usa base 1.

Wavelet tree — **Complexity:** $O(N \log \sigma build, \log \sigma query)$

Usage: 1. `WT(int n)` → inicializa la estructura para n elementos. 2. `add(int v)` → agrega un valor comprimido al final. 3. `range(int k, int s, int e, int a, int b)` → cuenta valores en [a, b) dentro de posiciones [s, e). *Note:* Comprimir valores antes de insertar.

STL extended set — **Complexity:** $O(\log Nperoperation)$

Usage: 1. `ordered_set` → set ordenado con orden estadistico. 2. `find_by_order(k)` → devuelve iterador al k-esimo elemento. 3. `order_of_key(x)` → cuenta elementos estrictamente menores que x. *Note:* Requiere GNU PBDS; no es C++ estandar puro.

STL rope — **Complexity:** $O(\log N per split/insert/erase)$
Usage: 1. `rope<T>` → secuencia balanceada con operaciones de edicion. 2. `insert(pos, value)` → inserta en una posicion. 3. `erase(pos, len)` → borra un rango contiguo. *Note:* Requiere extension GNU __gnu_cxx::rope.

Treap as BST — **Complexity:** $O(\log N expected per operation)$
Usage: 1. `split(pitem t, int key, pitem& l, pitem& r)` → separa por clave: l ¡ key, r ¿= key. 2. `insert(pitem& t, pitem it)` → inserta un nodo. 3. `kth(pitem t, int k)` → devuelve el nodo k-esimo en orden. *Note:* Aleatorizado; crear nodos con prioridad random.

Implicit treap — **Complexity:** $O(\log N expected per operation)$
Usage: 1. `merge(pitem& t, pitem l, pitem r)` → concatena dos treaps. 2. `split(pitem t, pitem& l, pitem& r, int sz)` → deja sz elementos en l. 3. `output(pitem t)` → recorre la secuencia en orden. *Note:* Las claves son posiciones implicitas; propagar lazy con push.

Implicit treap with father — **Complexity:** $O(\log N expected per operation)$
Usage: 1. `merge(pitem& t, pitem l, pitem r)` → concatena y actualiza padres. 2. `split(pitem t, pitem& l, pitem& r, int sz)` → separa por tamano. 3. `root(pitem t, int& pos)` → devuelve raiz y posicion actual del nodo. *Note:* Util cuando se conservan punteros a nodos individuales.

Link-cut tree — **Complexity:** $O(\log N amortized per operation)$
Usage: 1. `evert(pitem v)` → convierte v en raiz del arbol representado. 2. `link(pitem v, pitem w)` → agrega arista haciendo v hijo de w. 3. `cut(pitem v, pitem w)` → elimina la arista entre v y w. *Note:* Antes de usar, inicializar campos de cada nodo.

Link-cut tree with path aggregate — **Complexity:** $O(\log N amortized per operation)$
Usage: 1. `mkR(Node x)` → hace a x raiz representada. 2. `link(Node x, Node y)` → conecta dos componentes. 3. `query(Node x, Node y)` → consulta el agregado del camino x-y. *Note:* Modificar mOp, qOp y deltas para el agregado requerido.

Static convex hull trick — **Complexity:** $O(\log N per query after monotone insert)$
Usage: 1. `CHT::add(Line l)` → agrega una recta al hull. 2. `CHT::eval(int p, tc x)` → evalua la recta p en x. 3. `CHT::query(tc x)` → devuelve el minimo en x. *Note:* Para maximo, cambiar signos como indica el comentario del archivo.

Dynamic convex hull trick — **Complexity:** $O(\log N per insert/query)$
Usage: 1. `HullDynamic::add_line(tc m, tc b)` → agrega y = m x + b. 2. `HullDynamic::eval(Line l, tc x)` → evalua una recta. 3. `HullDynamic::query(tc x)` → devuelve el maximo en x. *Note:* Implementado para maximo.

Li Chao tree — **Complexity:** $O(\log C per insert/query)$
Usage: 1. `LiChaoTree(T xmin, T xmax)` → crea el dominio discreto/entero [xmin, xmax). 2. `add_line(Line nw)` → inserta una recta. 3. `query(T x)` → devuelve el minimo en x. *Note:* Ajustar INF y dominio al rango de coordenadas.

Gain-cost set — **Complexity:** $O(\log N amortized per insert/query)$
Usage: 1. `GCS::add(ii x)` → agrega un par y elimina dominados. 2. `GCS::query(int v)` → calcula el mejor valor para v segun los pares activos. *Note:* Mantiene frontera no dominada en un set.

Disjoint intervals — **Complexity:** $O(K \log N per insertion)$
Usage: 1. `disjoint_intervals::add(ii v)` → agrega el intervalo y une solapados. 2. `set<ii> s` → contiene intervalos disjuntos ordenados. *Note:* Intervalos semiarbitrarios segun ii; revisar inclusividad antes de copiar.

1.2 Graphs

Topological sort — **Complexity:** $O(V + E)$
Usage: 1. `tsort()` → devuelve un orden topologico lexicograficamente minimo. 2. `priority_queue<int, vector<int>, greater<int>>` → elige siempre el nodo libre mas chico. *Note:* Si el resultado tiene menos de n nodos, hay ciclo.

Kruskal MST — **Complexity:** $O(E \log E)$
Usage: 1. `uf_init()` → inicializa union-find. 2. `uf_join(int x, int y)` → une componentes si eran distintas. 3. `kruskal()` → devuelve el peso del MST asumiendo grafo conexo. *Note:* Ordenar aristas por peso antes de procesarlas si el archivo no lo hace por ti.

Dijkstra — **Complexity:** $O((V + E) \log V)$
Usage: 1. `dijkstra(int x)` → calcula distancias minimas desde x. 2. `dist[v]` → queda con -1 si v no es alcanzable. *Note:* Requiere pesos no negativos.

Bellman-Ford — **Complexity:** $O(VE)$
Usage: 1. `bford(int src)` → calcula distancias desde src y propaga ciclos negativos. 2. `d[v]` → guarda INF, distancia finita o -INF segun alcanzabilidad. *Note:* Sirve con pesos negativos; detecta nodos afectados por ciclos negativos alcanzables.

Floyd-Warshall — **Complexity:** $O(V^3)$
Usage: 1. `floyd()` → reemplaza g por distancias minimas entre todos los pares. 2. `inNegCycle(int v)` → indica si v esta en un ciclo negativo. 3. `hasNegCycle(int a, int b)` → indica si hay ciclo negativo en algun camino a-b. *Note:* Inicializar g[i][i]=0 y g[u][v]=peso.

SCC and 2-SAT — **Complexity:** $O(V + E)$
Usage: 1. `scc()` → calcula componentes fuertemente conexas. 2. `addor(int a, int b)` → agrega la clausula a OR b. 3. `satisf(int nvar)` → devuelve si la instancia 2-SAT es satisfacible. *Note:* Usa literales codificados con neg(a).

Articulation, bridges and biconnected — **Complexity:** $O(V + E)$
Usage: 1. `add_edge(int u, int v)` → agrega una arista no dirigida. 2. `doit()` → ejecuta DFS y marca puentes/componentes. 3. `edge.bridge` → queda verdadero para aristas puente. *Note:* Reinicializar vectores globales entre casos.

Chu-Liu Edmonds arborescence — **Complexity:** $O(VE)$
Usage: 1. `ChuLiu(int n, int r)` → crea instancia con n nodos y raiz r. 2. `add_edge(int u, int v, tw w)` → agrega arco dirigido u -¿ v. 3. `doit()` → devuelve el costo de la arborescencia minima. *Note:* Los nodos deben ser alcanzables desde la raiz.

LCA binary lifting — **Complexity:** $O(N \log Nbuild, \log Nquery)$
Usage: 1. `lca_init()` → precalcula profundidad y tabla binaria. 2. `lca(int x, int y)` → devuelve el ancestro comun mas bajo. *Note:* Asume arbol enraizado en 0.

Heavy-light decomposition — **Complexity:** $O(Nbuild, \log^2 Nquery)$
Usage: 1. `hld_init()` → calcula cadenas y posiciones. 2. `query(int x, int y, STree& rmq)` → consulta el camino x-y usando segment tree. *Note:* Combinar con un segment tree sobre las posiciones linealizadas.

Centroid decomposition — **Complexity:** $O(N \log N)$
Usage: 1. `centroid()` → construye la descomposicion. 2. `cdfs(int x, int f, int sz)` → procesa recursivamente centroides. *Note:* Completar la logica de actualizacion/consulta dentro del DFS segun el problema.

Parallel DFS — **Complexity:** $O(V + E)$
Usage: 1. `Tree` → estructura auxiliar para DFS en paralelo. 2. `Tree::go(int x)` → recorre el arbol manteniendo estado compartido. *Note:* Snippet especializado; revisar invariantes del problema antes de usar.

Eulerian path — **Complexity:** $O(V + E)$
Usage: 1. `add_edge(int a, int b)` → agrega arista al multigrafo. 2. `get_path(int x)` → devuelve un camino euleriano que empieza en x. *Note:* Verificar grados/conectividad antes de llamar si el problema no lo garantiza.

Offline dynamic connectivity — **Complexity:** $O((N + Q) \log Q)$
Usage: 1. `DynCon(int n, int q)` → prepara estructura offline. 2. `add(int l, int r, Query e)` → activa una arista en el intervalo [l, r). 3. `go(int k, int s, int e)` → resuelve recursivamente consultas con rollback. *Note:* Procesa consultas offline con union-find rollback.

Dominator tree — **Complexity:** $O((V + E)\alpha(V))$
Usage: 1. `dominators(int rt)` → calcula dominadores desde la raiz rt. 2. `idom[v]` → padre inmediato de v en el arbol de dominadores. *Note:* Algoritmo Lengauer-Tarjan; nodos no alcanzables requieren filtrado.

Edmonds blossom — **Complexity:** $O(V^2E)$
Usage: 1. `edmonds()` → devuelve tamano de matching maximo en grafo general. 2. `aug(int s, int t)` → busca camino aumentante desde s. *Note:* Grafo no dirigido; limpiar matching global entre casos.

1.3 Math

Integer floor division — **Complexity:** $O(1)$
Usage: 1. `floordiv(ll x, ll y, ll& q, ll& r)` → calcula q=floor(x/y) y residuo r. 2. `q, r` → satisfacen x = q*y + r con r ajustado al signo esperado. *Note:* Pensado para x negativo; revisar convencion de residuo si y ¡ 0.

Extended Euclid — **Complexity:** $O(\log min(a, b))$
Usage: 1. `euclid(ll a, ll b, ll& x, ll& y)` → devuelve gcd(a,b) y coeficientes x,y. 2. `x, y` → cumplen a*x + b*y = gcd(a,b).

Sieve of Eratosthenes — **Complexity:** $O(N \log \log N)$
Usage: 1. `init_sieve()` → precalcula primalidad/factor menor hasta MAXN. 2. `sv[x]` → indica factor o estado de x segun el snippet. *Note:* Ajustar MAXN antes de compilar.

Generate divisors — **Complexity:** $O(d(n))$
Usage: 1. `divisors(vector<pair<ll,int>> f)` → genera todos los divisores desde la factorizacion. 2. `div_rec(...)` → recursion interna para multiplicar potencias. *Note:* La entrada es factorizacion primo, exponente.

Pollard Rho factorization — **Complexity:** $O(probabilistic, aboutn^{1/4})$
Usage: 1. `rabin(ll n)` → test probabilistico/deterministico practico de primalidad. 2. `rho(ll n)` → encuentra un factor no trivial. 3. `fact(ll n, map<ll,int>& f)` → factoriza n en el mapa f. *Note:* Para la version con primos chicos, llamar init_sv primero.

Simpson integration — **Complexity:** $O(N)$
Usage: 1. `integrate(double f(double), double a, double b, int n)` → aproxima la integral de f en [a,b]. 2. `n` → cantidad de subintervalos; mayor n mejora precision. *Note:* Usar n par/grande para mejor estabilidad.

Polynomial utilities — **Complexity:** $O(dependsonoperation)$
Usage: 1. `poly<T>` → representa polinomios por coeficientes. 2. `roots(poly<> p)` → busca raices enteras. 3. `interpolate(vector<tp> x, vector<tp> y)` → construye polinomio interpolante. *Note:* Complejidades son cuadraticas en las operaciones basicas incluidas.

Bairstow roots — **Complexity:** $O(ITER * degree)$
Usage: 1. `bairstow(poly<> p)` → aproxima un factor cuadratico de p. 2. `roots(poly<> p)` → devuelve raices reales aproximadas. *Note:* Requiere el tipo poly del archivo de polinomios.

Fast Fourier Transform — **Complexity:** $O(N \log N)$
Usage: 1. `dft(CD* a, int n, bool inv)` → transforma in-place. 2.

`multiply(poly& p1, poly& p2)` → multiplica polinomios/convoluciona vectores. *Note:* Elegir version FFT o NTT segun comentarios del archivo.

FFT polynomial operations — Complexity: $O(N \log N perhigh - leveloperation)$

Usage: 1. `invert(poly& p, int d)` → inverso formal modulo x^d. 2. `log(poly& p, int d)` → logaritmo formal modulo x^d. 3. `interpolate(vector<tf>& x, vector<tf>& y)` → interpola puntos usando arbol de productos. *Note:* Depende de multiply y aritmetica modular definida en el notebook.

Fast Hadamard Transform — Complexity: $O(N \log N)$

Usage: 1. `fht(ll* p, int n, bool inv)` → transforma para XOR/AND/OR segun variante. 2. `multiply(vector<ll>& p1, vector<ll>& p2)` → convoluciona con la transformada elegida. *Note:* Ajustar el operador dentro del archivo segun XOR, AND u OR.

Karatsuba multiplication — Complexity: $O(N^{1.585})$

Usage: 1. `karatsura(int n, tp* p, tp* q, tp* r)` → multiplica arreglos de tamano n. 2. `multiply(vector<tp> p0, vector<tp> p1)` → devuelve la convolucion. *Note:* El nombre de la funcion interna esta escrito como en el snippet.

Linear Diophantine equations — Complexity: $O(\log min(a, b))$

Usage: 1. `extendedEuclid(ll a, ll b)` → devuelve coeficientes de Bezout. 2. `diophantine(ll a, ll b, ll r)` → resuelve a*x + b*y = r y direccion de soluciones. *Note:* r debe ser multiplo de gcd(a,b).

Modular inverse — Complexity: $O(\log mod)$

Usage: 1. `inv(ll a, ll mod)` → devuelve el inverso de a modulo mod. *Note:* Requiere gcd(a, mod)=1.

Chinese remainder theorem — Complexity: $O(K \log M)$

Usage: 1. `sol(tuple<ll,ll,ll> c)` → resuelve una congruencia a*x = b mod m. 2. `crt(vector<tuple<ll,ll,ll>> cond)` → combina condiciones y devuelve solucion, lcm. *Note:* Requiere inv, diophantine, gcd y que el lcm quepa en ll.

Mobius function — Complexity: $O(N \log N)$

Usage: 1. `mobius()` → precalcula mu hasta MAXN. 2. `mu[i]` → queda con la funcion de Mobius de i. *Note:* Ajustar MAXN y limpiar arreglos entre usos.

Matrix fast exponentiation — Complexity: $O(N^3 \log E)$

Usage: 1. `ones(int n)` → devuelve matriz identidad n x n. 2. `be(Matrix b, ll e)` → calcula b^e por exponenciacion binaria. *Note:* Cambiar multiplicacion/modulo dentro del operador si hace falta.

Gaussian elimination — Complexity: $O(NMmin(N, M))$

Usage: 1. `reduce(vector<vector<double>>& x)` → reduce la matriz y devuelve determinante cuando aplica. 2. `x` → queda en forma escalonada reducida. *Note:* Usa double y EPS; para modulares cambiar division y comparacion.

Simplex — Complexity: $O(exponentialworst - case, fastinpractice)$

Usage: 1. `simplex(int n, int m)` → resuelve LP en forma estandar. 2. `pivot(int x, int y)` → realiza pivoteo interno. *Note:* Revisar convencion de restricciones antes de copiar.

Discrete logarithm — Complexity: $O(\sqrt{m})$

Usage: 1. `discrete_log(ll a, ll b, ll m)` → encuentra x tal que $a^x \equiv b \pmod m$. *Note:* Baby-step giant-step; revisar casos gcd(a,m) != 1.

Berlekamp-Massey — Complexity: $O(N^2)$

Usage: 1. `BM(vi x)` → devuelve coeficientes de la recurrencia minima. 2. `cur` → coeficientes normalizados modulo MOD. *Note:* La secuencia y MOD son globales/externos.

Linear recurrence — Complexity: $O(K^2 \log N)$

Usage: 1. `LinearRec(vector<int> terms, vector<int> trans)` → define terminos iniciales y transicion. 2. `calc(ll k)` → devuelve el k-esimo termino. *Note:* Combina Berlekamp-Massey con exponenciacion de polinomios.

Tonelli-Shanks — Complexity: $O(\log^2 p)$

Usage: 1. `legendre(ll a, ll p)` → calcula simbolo de Legendre. 2. `tonelli_shanks(ll n, ll p)` → devuelve una raiz cuadrada de n modulo p. *Note:* p debe ser primo.

Prime counting — Complexity: $O(n^{2/3}/\log n)$

Usage: 1. `prime_count(ll n)` → devuelve pi(n), cantidad de primos ¡= n. *Note:* Usar ll; revisar memoria para n grande.

Number Theoretic Transform — Complexity: $O(N \log N)$

Usage: 1. `FFT<u, uu, p, root>` → transformada modular parametrizada. 2. `conv_small(const poly& a, const poly& b)` → convolucion bajo modulo NTT. 3. `conv_sunzi(const poly& a, const poly& b, ll m)` → convolucion con CRT para modulo arbitrario. *Note:* Elegir primos/raices compatibles con el tamano.

Lattice points under line — Complexity: $O(\log max(a, b, c))$

Usage: 1. `f(ll a, ll b, ll c)` → suma pisos relacionada con ax+b sobre c. 2. `g(ll a, ll b, ll c, ll X, ll Y)` → cuenta puntos bajo una recta en una caja. *Note:* Verificar inclusividad de bordes para el problema.

1.4 Geometry

Point — Complexity: $O(1 peroperation)$

Usage: 1. `pt` → punto/vector 2D con operaciones aritmeticas. 2. `pt::norm()` → devuelve longitud. 3. `pt::rot(pt r)` → rota usando el vector r. *Note:* Para 3D, agregar coordenada z como indica el comentario.

Line — Complexity: $O(1 peroperation)$

Usage: 1. `ln(pt p, pt q)` → crea recta por dos puntos. 2. `ln::proj(pt p)` →

proyecta p sobre la recta. 3. 3. `operator^(ln l, ln m)` → interseccion de dos rectas. *Note:* Depende de pt y EPS.

Circle — Complexity: $O(1 perprimitive, N forintercircles)$
Usage: 1. `circle(pt o, double r)` → define centro y radio. 2. `circle::has(pt p)` → clasifica punto respecto al circulo. 3. `intercircles(vector<circle> c)` → calcula area cubierta por circulos. *Note:* Usa EPS; revisar tangencias.

Polygon — Complexity: $O(N perbasicquery, \log^2 N dynamichullquery)$
Usage: 1. `pol::area()` → devuelve area orientada. 2. `pol::has(pt q)` → clasifica punto respecto del poligono. 3. `add(pt q) / query(pt v)` → mantiene hull dinamico y consulta maximo dot. *Note:* Convenciones de orientacion y borde dependen de sgn/EPS.

Plane — Complexity: $O(1 peroperation)$
Usage: 1. `plane(pt p, pt n)` → plano por punto y normal. 2. `plane::side(pt q)` → clasifica q respecto al plano. 3. `operator^(plane a, plane b)` → interseccion de planos cuando aplica. *Note:* Pensado para geometria 3D con pt extendido.

Radial order — Complexity: $O(N \log N sort)$
Usage: 1. `Cmp` → comparador angular para ordenar puntos radialmente. 2. `sort(v.begin(), v.end(), Cmp())` → ordena por semiplano y producto cruz. *Note:* El comentario del archivo pide const en operator- de pt.

Convex hull — Complexity: $O(N \log N)$
Usage: 1. `chull(vector<pt> p)` → devuelve el hull convexo de los puntos. *Note:* Revisar si conserva o elimina colineales segun la condicion de giro.

Planar graph dual — Complexity: $O(E \log E)$
Usage: 1. `get_dual(vector<pt> p)` → construye caras/dual desde embedding planar. 2. `g[u]` → lista de adyacencia geometrica del grafo primal. *Note:* Los puntos deben corresponder a nodos del embedding sin cruces.

Halfplane intersection — Complexity: $O(N \log N)$
Usage: 1. `halfplane(pt p, pt q)` → define semiplano por borde orientado. 2. `intersect(vector<halfplane> b)` → devuelve poligono de interseccion. *Note:* Agregar caja grande si la interseccion puede ser no acotada.

KD tree — Complexity: $O(N \log N build, about \log N queryaverage)$
Usage: 1. `KDTree(vector<pt> p)` → construye KD-tree. 2. `nearest(pt q)` → consulta punto mas cercano. *Note:* Rendimiento depende de distribucion de puntos.

Minkowski sum — Complexity: $O(N + M after sorting)$
Usage: 1. `reorder(vector<pt>& p)` → rota poligono para empezar por el menor punto. 2. `minkowski_sum(vector<pt> p, vector<pt> q)` → devuelve suma de Minkowski de poligonos convexos. *Note:* Los poligonos deben estar en orden alrededor del borde.

Delaunay triangulation — Complexity: $O(N \log N expected)$
Usage: 1. `delaunay(vector<pt> v)` → devuelve triangulacion como adyacencias. 2. `in_c(pt a, pt b, pt c, pt p)` → test de circuncirculo interno. *Note:* Evitar puntos duplicados; sensible a EPS/orientacion.

1.5 Strings

KMP — Complexity: $O(N + M)$
Usage: 1. `kmppre(string& t)` → calcula funcion de prefijo/bordes del patron. 2. `kmp(string& s, string& t)` → encuentra ocurrencias de t en s. *Note:* Adaptar el cuerpo de kmp para guardar/imprimir matches.

Z function — Complexity: $O(N)$
Usage: 1. `z_function(string& s)` → devuelve z[i], largo del prefijo que empieza en i. *Note:* Util para matching concatenando patron + separador + texto.

Manacher — Complexity: $O(N)$
Usage: 1. `manacher(string& s)` → precalcula radios de palindromos pares e impares. 2. `d1, d2` → arreglos de radios segun paridad. *Note:* Revisar convencion de indices del snippet.

Aho-Corasick — Complexity: $O(total length + matches)$
Usage: 1. `aho_init()` → inicializa el automata. 2. `add_string(string s, int id)` → agrega patron con identificador. 3. `go(int v, char c)` → transicion del automata con links calculados lazy. *Note:* No olvidar llamar aho_init antes de insertar.

Suffix automaton — Complexity: $O(N)$
Usage: 1. `sa_init()` → inicializa automata vacio. 2. `sa_extend(char c)` → agrega un caracter al final. 3. `state::next` → transiciones por caracter. *Note:* Limpiar next en cada estado nuevo.

Palindromic tree — Complexity: $O(N)$
Usage: 1. `palindromic_tree` → estructura Eertree para palindromos distintos. 2. `add(int c)` → agrega un caracter y actualiza sufijo palindromico. 3. `node` → almacena longitud, link y transiciones. *Note:* Usar alfabeto/offset adecuados.

Suffix array slow — Complexity: $O(N \log^2 N)$
Usage: 1. `constructSA(string& s)` → devuelve suffix array usando comparacion de substrings. 2. `sacomp(int lhs, int rhs)` → comparador de sufijos. *Note:* Simple pero lento; preferir la version O(N log N) para strings grandes.

Suffix array — Complexity: $O(N \log N)$
Usage: 1. `constructSA(string& s)` → construye suffix array. 2. `csort(vector<int>& sa, vector<int>& r, int k)` → counting sort por ranking desplazado. *Note:* Agregar sentinel si el problema lo requiere.

LCP array — Complexity: $O(N)$
Usage: 1. `computeLCP(string& s, vector<int>& sa)` → devuelve LCP entre sufijos consecutivos en sa. *Note:* Requiere suffix array valido de s.

Suffix tree — **Complexity:** $O(N)$

Usage: 1. `SuffixTree(string s)` → construye arbol de sufijos con Ukkonen. 2. `go(...)` → navega por transiciones internas. 3. `split(...)` → divide una arista durante la extension. *Note:* Agregar sentinel unico al string para hojas explicitas.

String hashing — **Complexity:** $O(Nbuild, 1query)$

Usage: 1. `Hash(string s)` → precalcula hashes/potencias. 2. `get(int l, int r)` → devuelve hash del substring [l, r). *Note:* Usar doble hash o modulo robusto si hay riesgo de colision.

String hashing with __int128 — **Complexity:** $O(Nbuild, 1query)$

Usage: 1. `Hash(string s)` → precalcula hash con multiplicacion amplia. 2. `get(int l, int r)` → devuelve hash de substring. *Note:* Usa __int128; requiere compilador GNU/Clang compatible.

1.6 Flow

Bipartite matching DFS — **Complexity:** $O(VE)$

Usage: 1. `match(int x)` → intenta aumentar desde el nodo izquierdo x. 2. `mm()` → devuelve las parejas encontradas. *Note:* Mas simple que Hopcroft-Karp; usar en grafos chicos.

Hopcroft-Karp — **Complexity:** $O(E\sqrt{V})$

Usage: 1. `bfs()` → construye capas desde libres izquierdos. 2. `dfs(int x)` → aumenta por caminos de capas. 3. `mm()` → devuelve el tamano del matching maximo. *Note:* Particiones izquierda/derecha definidas por los arreglos globales.

Hungarian algorithm — **Complexity:** $O(N^3)$

Usage: 1. `Hungarian(int n, int m)` → resuelve asignacion rectangular. 2. `assign()` → devuelve costo y matching optimo. 3. `L[i]` → columna asignada a la fila i. *Note:* Cambiar signo de costos para maximizacion.

Dinic max flow — **Complexity:** $O(V^2E)$

Usage: 1. `Dinic(int nodes)` → crea red con nodes vertices. 2. `add_edge(int s, int t, ll cap)` → agrega arco con capacidad. 3. `max_flow(int src, int dst)` → devuelve flujo maximo src-dst. *Note:* Despues del flujo, dist permite recuperar corte minimo como indica el comentario.

Min-cost max-flow — **Complexity:** $O(FE \log V)$

Usage: 1. `MCF(int n)` → crea red con n vertices. 2. `add_edge(int s, int t, tf cap, tc cost)` → agrega arco capacitado con costo. 3. `get_flow(int s, int t)` → devuelve {flujo, costo} maximo de costo minimo. *Note:* Usa potenciales y Dijkstra; costos negativos requieren potencial inicial valido.

1.7 Other

Mo algorithm — **Complexity:** $O((N+Q)\sqrt{N})$

Usage: 1. `mos()` → ordena consultas y mantiene ventana activa. 2. `add(pos) / remove(pos)` → actualizan la respuesta al mover extremos. 3. `qu{l,r,id}` → representa consulta offline. *Note:* Completar add/remove segun la funcion objetivo.

Divide and conquer DP optimization — **Complexity:** $O(KN \log N)$

Usage: 1. `doit(int s, int e, int s0, int e0, int i)` → calcula una capa DP restringiendo optimo. 2. `doall()` → ejecuta todas las capas necesarias. *Note:* Valido si el optimo es monotono.

Date conversion — **Complexity:** $O(1)$

Usage: 1. `dateToInt(int y, int m, int d)` → convierte fecha gregoriana a entero juliano. 2. `intToDate(int jd, int& y, int& m, int& d)` → convierte entero a fecha. 3. `DayOfWeek(int d, int m, int y)` → devuelve dia de semana empezando en domingo. *Note:* Revisar calendario esperado por el problema.

2 Data structures

2.1 Segment tree

```
#define oper min
#define NEUT INF
struct STree { // segment tree for min over integers
  vector<int> st;int n;
  STree(int n): st(4*n+5,NEUT), n(n) {}
  void init(int k, int s, int e, int *a){
    if(s+1==e){st[k]=a[s];return;}
    int m=(s+e)/2;
    init(2*k,s,m,a);init(2*k+1,m,e,a);
    st[k]=oper(st[2*k],st[2*k+1]);
  }
  void upd(int k, int s, int e, int p, int v){
    if(s+1==e){st[k]=v;return;}
    int m=(s+e)/2;
    if(p<m)upd(2*k,s,m,p,v);
    else upd(2*k+1,m,e,p,v);
    st[k]=oper(st[2*k],st[2*k+1]);
  }
  int query(int k, int s, int e, int a, int b){
    if(s>=b||e<=a)return NEUT;
    if(s>=a&&e<=b)return st[k];
    int m=(s+e)/2;
    return oper(query(2*k,s,m,a,b),query(2*k+1,m,e,a,b));
  }
  void init(int *a){init(1,0,n,a);}
  void upd(int p, int v){upd(1,0,n,p,v);}
  int query(int a, int b){return query(1,0,n,a,b);}
}; // usage: STree rmq(n);rmq.init(x);rmq.upd(i,v);rmq.query(s,e);
```

2.2 Segment tree - Lazy propagation

```
struct STree { // example: range sum with range addition
  vector<int> st,lazy;int n;
  STree(int n): st(4*n+5,0), lazy(4*n+5,0), n(n) {}
  void init(int k, int s, int e, int *a){
    lazy[k]=0;  // lazy neutral element
    if(s+1==e){st[k]=a[s];return;}
    int m=(s+e)/2;
    init(2*k,s,m,a);init(2*k+1,m,e,a);
    st[k]=st[2*k]+st[2*k+1]; // operation
  }
  void push(int k, int s, int e){
    if(!lazy[k])return; // if neutral, nothing to do
    st[k]+=(e-s)*lazy[k]; // update st according to lazy
    if(s+1<e){ // propagate to children
      lazy[2*k]+=lazy[k];
      lazy[2*k+1]+=lazy[k];
    }
    lazy[k]=0; // clear node lazy
  }
  void upd(int k, int s, int e, int a, int b, int v){
    push(k,s,e);
    if(s>=b||e<=a)return;
    if(s>=a&&e<=b){
      lazy[k]+=v; // accumulate lazy
      push(k,s,e);return;
    }
    int m=(s+e)/2;
    upd(2*k,s,m,a,b,v);upd(2*k+1,m,e,a,b,v);
    st[k]=st[2*k]+st[2*k+1]; // operation
  }
  int query(int k, int s, int e, int a, int b){
    if(s>=b||e<=a)return 0; // operation neutral
    push(k,s,e);
    if(s>=a&&e<=b)return st[k];
    int m=(s+e)/2;
    return query(2*k,s,m,a,b)+query(2*k+1,m,e,a,b); // operation
  }
  void init(int *a){init(1,0,n,a);}
  void upd(int a, int b, int v){upd(1,0,n,a,b,v);}
  int query(int a, int b){return query(1,0,n,a,b);}
}; // usage: STree rmq(n);rmq.init(x);rmq.upd(s,e,v);rmq.query(s,e);
```

2.3 Segment tree - Persistence

```
#define oper(a,b) min(a,b)
#define NEUT INF
struct STree { // persistent segment tree for min over integers
  vector<int> st, L, R; int n,sz,rt;
  STree(int n): st(1,NEUT),L(1,0),R(1,0),n(n),rt(0),sz(1){}
  int new_node(int v, int l=0, int r=0){
    int ks=SZ(st);
```

```
    st.push_back(v);L.push_back(l);R.push_back(r);
    return ks;
  }
  int init(int s, int e, int *a){ // not necessary in most cases
    if(s+1==e)return new_node(a[s]);
    int m=(s+e)/2,l=init(s,m,a),r=init(m,e,a);
    return new_node(oper(st[l],st[r]),l,r);
  }
  int upd(int k, int s, int e, int p, int v){
    int ks=new_node(st[k],L[k],R[k]);
    if(s+1==e){st[ks]=v;return ks;}
    int m=(s+e)/2,ps;
    if(p<m)ps=upd(L[ks],s,m,p,v),L[ks]=ps;
    else ps=upd(R[ks],m,e,p,v),R[ks]=ps;
    st[ks]=oper(st[L[ks]],st[R[ks]]);
    return ks;
  }
  int query(int k, int s, int e, int a, int b){
    if(e<=a||b<=s)return NEUT;
    if(a<=s&&e<=b)return st[k];
    int m=(s+e)/2;
    return oper(query(L[k],s,m,a,b),query(R[k],m,e,a,b));
  }
  int init(int *a){return init(0,n,a);}
  int upd(int k, int p, int v){return rt=upd(k,0,n,p,v);}
  int upd(int p, int v){return upd(rt,p,v);} // update on last root
  int query(int k,int a, int b){return query(k,0,n,a,b);};
}; // usage: STree rmq(n);root=rmq.init(x);new_root=rmq.upd(root,i,v);
    rmq.query(root,s,e);
```

2.4 Segment tree - 2D

```
int n,m;
int a[MAXN][MAXN],st[2*MAXN][2*MAXN];
void build(){
  for (int i = 0; i < n; ++i)for (int j = 0; j < m; ++j)st[i+n][j+m]=a[i
      ][j];
  for (int i = 0; i < n; ++i)for(int j=m-1;j;--j)
    st[i+n][j]=op(st[i+n][j<<1],st[i+n][j<<1|1]);
  for(int i=n-1;i;--i)for (int j = 0; j < 2*m; ++j)
    st[i][j]=op(st[i<<1][j],st[i<<1|1][j]);
}
void upd(int x, int y, int v){
  st[x+n][y+m]=v;
  for(int j=y+m;j>1;j>>=1)st[x+n][j>>1]=op(st[x+n][j],st[x+n][j^1]);
  for(int i=x+n;i>1;i>>=1)for(int j=y+m;j;j>>=1)
    st[i>>1][j]=op(st[i][j],st[i^1][j]);
}
int query(int x0, int x1, int y0, int y1){
  int r=NEUT;
  for(int i0=x0+n,i1=x1+n;i0<i1;i0>>=1,i1>>=1){
    int t[4],q=0;
    if(i0&1)t[q++]=i0++;
    if(i1&1)t[q++]=--i1;
    for (int k = 0; k < q; ++k)for(int j0=y0+m,j1=y1+m;j0<j1;j0>>=1,j1
        >>=1){
      if(j0&1)r=op(r,st[t[k]][j0++]);
      if(j1&1)r=op(r,st[t[k]][--j1]);
    }
  }
  return r;
}
```

2.5 Sparse table (static RMQ)

```
#define oper min
int st[K][1<<K];int n;  // K such that 2^K>n
void st_init(int *a){
  for (int i = 0; i < n; ++i)st[0][i]=a[i];
  for (int k = 1; k < K; ++k)for (int i = 0; i < n-(1<<k)+1; ++i)
    st[k][i]=oper(st[k-1][i],st[k-1][i+(1<<(k-1))]);
}
int st_query(int s, int e){
  int k=31-__builtin_clz(e-s);
  return oper(st[k][s],st[k][e-(1<<k)]);
}
```

2.6 Fenwick tree

```
int ft[MAXN+1]; // for more dimensions, make ft multi-dimensional
void upd(int i0, int v){ // add v to i0th element (0-based)
  // add extra fors for more dimensions
  for(int i=i0+1;i<=MAXN;i+=i&-i)ft[i]+=v;
}
int get(int i0){ // get sum of range [0,i0)
  int r=0;
  // add extra fors for more dimensions
```

```
  for(int i=i0;i;i-=i&-i)r+=ft[i];
  return r;
}
int get_sum(int i0, int i1){ // get sum of range [i0,i1) (0-based)
  return get(i1)-get(i0);
}
```

2.7 Wavelet tree

```
struct WT {
  vector<int> wt[1<<20];int n;
  void init(int k, int s, int e){
    if(s+1==e)return;
    wt[k].clear();wt[k].push_back(0);
    int m=(s+e)/2;
    init(2*k,s,m);init(2*k+1,m,e);
  }
  void add(int k, int s, int e, int v){
    if(s+1==e)return;
    int m=(s+e)/2;
    if(v<m)wt[k].push_back(wt[k].back()),add(2*k,s,m,v);
    else wt[k].push_back(wt[k].back()+1),add(2*k+1,m,e,v);
  }
  int query0(int k, int s, int e, int a, int b, int i){
    if(s+1==e)return s;
    int m=(s+e)/2;
    int q=(b-a)-(wt[k][b]-wt[k][a]);
    if(i<q)return query0(2*k,s,m,a-wt[k][a],b-wt[k][b],i);
    else return query0(2*k+1,m,e,wt[k][a],wt[k][b],i-q);
  }
  void upd(int k, int s, int e, int i){
    if(s+1==e)return;
    int m=(s+e)/2;
    int v0=wt[k][i+1]-wt[k][i],v1=wt[k][i+2]-wt[k][i+1];
    if(!v0&&!v1)upd(2*k,s,m,i-wt[k][i]);
    else if(v0&&v1)upd(2*k+1,m,e,wt[k][i]);
    else if(v0)wt[k][i+1]--;
    else wt[k][i+1]++;
  }
  void init(int _n){n=_n;init(1,0,n);} // (values in range [0,n))
  void add(int v){add(1,0,n,v);}
  int query0(int a, int b, int i){ // ith element in range [a,b)
    return query0(1,0,n,a,b,i);    // (if it was sorted)
  }
  void upd(int i){ // swap positions i,i+1
    upd(1,0,n,i);
  }
};
```

2.8 STL extended set

```
#include<ext/pb_ds/assoc_container.hpp>
#include<ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
typedef tree<int,null_type,less<int>,rb_tree_tag,
    tree_order_statistics_node_update> ordered_set;
// find_by_order(i) -> iterator to ith element
// order_of_key(k) -> position (int) of lower_bound of k
```

2.9 STL rope

```
#include <ext/rope>
using namespace __gnu_cxx;
rope<int> s;
// Sequence with O(log(n)) random access, insert, erase at any position
// s.push_back(x);
// s.insert(i,r) // insert rope r at position i
// s.erase(i,k) // erase subsequence [i,i+k)
// s.substr(i,k) // return new rope corresponding to subsequence [i,i+k)
// s[i] // access ith element (cannot modify)
// s.mutable_reference_at(i) // acces ith element (allows modification)
// s.begin() and s.end() are const iterators (use mutable_begin(),
    mutable_end() to allow modification)
```

2.10 Treap (as BST)

```
typedef struct item *pitem;
struct item {
  int pr,key,cnt;
  pitem l,r;
  item(int key):key(key),pr(rand()),cnt(1),l(0),r(0) {}
};
int cnt(pitem t){return t?t->cnt:0;}
void upd_cnt(pitem t){if(t)t->cnt=cnt(t->l)+cnt(t->r)+1;}
void split(pitem t, int key, pitem& l, pitem& r){ // l: < key, r: >= key
  if(!t)l=r=0;
  else if(key<t->key)split(t->l,key,l,t->l),r=t;
```

```
  else split(t->r,key,t->r,r),l=t;
  upd_cnt(t);
}
void insert(pitem& t, pitem it){
  if(!t)t=it;
  else if(it->pr>t->pr)split(t,it->key,it->l,it->r),t=it;
  else insert(it->key<t->key?t->l:t->r,it);
  upd_cnt(t);
}
void merge(pitem& t, pitem l, pitem r){
  if(!l||!r)t=l?l:r;
  else if(l->pr>r->pr)merge(l->r,l->r,r),t=l;
  else merge(r->l,l,r->l),t=r;
  upd_cnt(t);
}
void erase(pitem& t, int key){
  if(t->key==key)merge(t,t->l,t->r);
  else erase(key<t->key?t->l:t->r,key);
  upd_cnt(t);
}
void unite(pitem &t, pitem l, pitem r){
  if(!l||!r){t=l?l:r;return;}
  if(l->pr<r->pr)swap(l,r);
  pitem p1,p2;split(r,l->key,p1,p2);
  unite(l->l,l->l,p1);unite(l->r,l->r,p2);
  t=l;upd_cnt(t);
}
pitem kth(pitem t, int k){
  if(!t)return 0;
  if(k==cnt(t->l))return t;
  return k<cnt(t->l)?kth(t->l,k):kth(t->r,k-cnt(t->l)-1);
}
pair<int,int> lb(pitem t, int key){ // position and value of lower_bound
  if(!t)return {0,1<<30}; // (special value)
  if(key>t->key){
    auto w=lb(t->r,key);w.first+=cnt(t->l)+1;return w;
  }
  auto w=lb(t->l,key);
  if(w.first==cnt(t->l))w.second=t->key;
  return w;
}
```

2.11 Treap (implicit key)

```
// example that supports range reverse and addition updates, and range
    sum query
// (commented parts are specific to this  problem)
typedef struct item *pitem;
struct item {
  int pr,cnt,val;
//  int sum; // (paramters for range query)
//  bool rev;int add; // (parameters for lazy prop)
  pitem l,r;
  item(int val): pr(rand()),cnt(1),val(val),l(0),r(0)/*,sum(val),rev(0),
      add(0)*/ {}
};
void push(pitem it){
  if(it){
    /*if(it->rev){
      swap(it->l,it->r);
      if(it->l)it->l->rev^=true;
      if(it->r)it->r->rev^=true;
      it->rev=false;
    }
    it->val+=it->add;it->sum+=it->cnt*it->add;
    if(it->l)it->l->add+=it->add;
    if(it->r)it->r->add+=it->add;
    it->add=0;*/
  }
}
int cnt(pitem t){return t?t->cnt:0;}
// int sum(pitem t){return t?push(t),t->sum:0;}
void upd_cnt(pitem t){
  if(t){
    t->cnt=cnt(t->l)+cnt(t->r)+1;
    // t->sum=t->val+sum(t->l)+sum(t->r);
  }
}
void merge(pitem& t, pitem l, pitem r){
  push(l);push(r);
  if(!l||!r)t=l?l:r;
  else if(l->pr>r->pr)merge(l->r,l->r,r),t=l;
  else merge(r->l,l,r->l),t=r;
  upd_cnt(t);
}
```

```
void split(pitem t, pitem& l, pitem& r, int sz){ // sz:desired size of l
  if(!t){l=r=0;return;}
  push(t);
  if(sz<=cnt(t->l))split(t->l,l,t->l,sz),r=t;
  else split(t->r,t->r,r,sz-1-cnt(t->l)),l=t;
  upd_cnt(t);
}
void output(pitem t){ // useful for debugging
  if(!t)return;
  push(t);
  output(t->l);printf(" %d",t->val);output(t->r);
}
// use merge and split for range updates and queries
```

2.12 Treap (with node father)

```
// node father is useful to keep track of the chain of each node
// alternative: splay tree
// IMPORTANT: add pointer f in struct item
void merge(pitem& t, pitem l, pitem r){
  push(l);push(r);
  if(!l||!r)t=l?l:r;
  else if(l->pr>r->pr)merge(l->r,l->r,r),l->r->f=t=l;
  else merge(r->l,l,r->l),r->l->f=t=r;
  upd_cnt(t);
}
void split(pitem t, pitem& l, pitem& r, int sz){
  if(!t){l=r=0;return;}
  push(t);
  if(sz<=cnt(t->l)){
    split(t->l,l,t->l,sz);r=t;
    if(l)l->f=0;
    if(t->l)t->l->f=t;
  }
  else {
    split(t->r,t->r,r,sz-1-cnt(t->l));l=t;
    if(r)r->f=0;
    if(t->r)t->r->f=t;
  }
  upd_cnt(t);
}
void push_all(pitem t){
  if(t->f)push_all(t->f);
  push(t);
}
pitem root(pitem t, int& pos){ // get root and position for node t
  push_all(t);
  pos=cnt(t->l);
  while(t->f){
    pitem f=t->f;
    if(t==f->r)pos+=cnt(f->l)+1;
    t=f;
  }
  return t;
}
```

2.13 Link-Cut tree

```
typedef struct item *pitem;
struct item {
  int pr;bool rev;
  pitem l,r,f,d;
  item():pr(rand()),l(0),r(0),f(0),d(0),rev(0){}
};
void push(pitem t){
  if(t&&t->rev){
    swap(t->l,t->r);
    if(t->l)t->l->rev^=1;
    if(t->r)t->r->rev^=1;
    t->rev=0;
  }
}
void merge(pitem& t, pitem l, pitem r){
  push(l);push(r);
  if(!l||!r)t=l?l:r;
  else if(l->pr>r->pr)merge(l->r,l->r,r),l->r->f=t=l;
  else merge(r->l,l,r->l),r->l->f=t=r;
}
void push_all(pitem t){
  if(t->f)push_all(t->f);
  push(t);
}
void split(pitem t, pitem& l, pitem& r){
  push_all(t);
  l=t->l;r=t->r;t->l=t->r=0;
  while(t->f){
```

```
    pitem f=t->f;t->f=0;
    if(t==f->l){
      if(r)r->f=f;
      f->l=r;r=f;
    }
    else {
      if(l)l->f=f;
      f->r=l;l=f;
    }
    t=f;
  }
  if(l)l->f=0;
  if(r)r->f=0;
}
pitem path(pitem p){return p->f?path(p->f):p;}
pitem tail(pitem p){push(p);return p->r?tail(p->r):p;}
pitem expose(pitem p){
  pitem q,r,t;
  split(p,q,r);
  if(q)tail(q)->d=p;
  merge(p,p,r);
  while(t=tail(p),t->d){
    pitem d=t->d;t->d=0;
    split(d,q,r);
    if(q)tail(q)->d=d;
    merge(p,p,d);merge(p,p,r);
  }
  return p;
}
pitem root(pitem v){return tail(expose(v));}
void evert(pitem v){expose(v)->rev^=1;v->d=0;}
void link(pitem v, pitem w){ // make v son of w
  evert(v);
  pitem p=path(v);
  merge(p,p,expose(w));
}
void cut(pitem v){ // cut v from its father
  pitem p,q;
  expose(v);split(v,p,q);v->d=0;
}
void cut(pitem v, pitem w){evert(w);cut(v);}
```

2.14 Convex hull trick (static)

```
typedef ll tc;
struct Line{tc m,h;};
struct CHT { // for minimum (for maximum just change the sign of lines)
  vector<Line> c;
  int pos=0;
  tc in(Line a, Line b){
    tc x=b.h-a.h,y=a.m-b.m;
    return x/y+(x%y?!((x>0)^(y>0)):0); // ==ceil(x/y)
  }
  void add(tc m, tc h){ // m's should be non increasing
    Line l=(Line){m,h};
    if(c.size()&&m==c.back().m){
      l.h=min(h,c.back().h);c.pop_back();if(pos)pos--;
    }
    while(c.size()>1&&in(c.back(),l)<=in(c[c.size()-2],c.back())){
      c.pop_back();if(pos)pos--;
    }
    c.push_back(l);
  }
  inline bool fbin(tc x, int m){return in(c[m],c[m+1])>x;}
  tc eval(tc x){
    // O(log n) query:
    int s=0,e=c.size();
    while(e-s>1){int m=(s+e)/2;
      if(fbin(x,m-1))e=m;
      else s=m;
    }
    return c[s].m*x+c[s].h;
    // O(1) query (for ordered x's):
    while(pos>0&&fbin(x,pos-1))pos--;
    while(pos<c.size()-1&&!fbin(x,pos))pos++;
    return c[pos].m*x+c[pos].h;
  }
};
```

2.15 Convex hull trick (dynamic)

```
typedef ll tc;
const tc is_query=-(1LL<<62); // special value for query
struct Line {
  tc m,b;
```

```
  mutable multiset<Line>::iterator it,end;
  const Line* succ(multiset<Line>::iterator it) const {
    return (++it==end? NULL : &*it);}
  bool operator<(const Line& rhs) const {
    if(rhs.b!=is_query)return m<rhs.m;
    const Line *s=succ(it);
    if(!s)return 0;
    return b-s->b<(s->m-m)*rhs.m;
  }
};
struct HullDynamic : public multiset<Line> { // for maximum
  bool bad(iterator y){
    iterator z=next(y);
    if(y==begin()){
      if(z==end())return false;
      return y->m==z->m&&y->b<=z->b;
    }
    iterator x=prev(y);
    if(z==end())return y->m==x->m&&y->b<=x->b;
    return (x->b-y->b)*(z->m-y->m)>=(y->b-z->b)*(y->m-x->m);
  }
  iterator next(iterator y){return ++y;}
  iterator prev(iterator y){return --y;}
  void add(tc m, tc b){
    iterator y=insert((Line){m,b});
    y->it=y;y->end=end();
    if(bad(y)){erase(y);return;}
    while(next(y)!=end()&&bad(next(y)))erase(next(y));
    while(y!=begin()&&bad(prev(y)))erase(prev(y));
  }
  tc eval(tc x){
    Line l=*lower_bound((Line){x,is_query});
    return l.m*x+l.b;
  }
};
```

2.16 Li-Chao tree

```
// LiChaoTree lct(sz) -> allows queries for all x in range [0,sz)
// lct.add(L)         -> add line L to range [0,sz) in O(log(sz))
// lct.add(L,st,en)   -> add line L to range [st,en) in O(log^2(sz))
// lct.query(x)       -> query maximum L(x) across all current lines in
    O(log(sz))

typedef long long T;
T INF=1e18;

struct Line{
  T a, b;
  Line():a(0),b(-INF){}
  Line(T a, T b):a(a),b(b){}
  T get(T x){return a*x+b;}
};

struct LiChaoTree{
  struct Node{
    Line line;
    Node *lc,*rc=0;
  }*root;
  int sz;
  LiChaoTree(int sz): root(new Node()), sz(sz){}

  void addRec(Node* &n, int tl, int tr, Line x){
    if(!n) n=new Node();
    if(n->line.get(tl)<x.get(tl)) swap(n->line,x);
    if(n->line.get(tr-1)>=x.get(tr-1)||tl+1==tr) return;
    int mid=(tl+tr)/2;
    if(n->line.get(mid)>x.get(mid)){
      addRec(n->rc,mid,tr,x);
    }
    else{
      swap(n->line,x);
      addRec(n->lc,tl,mid,x);
    }
  }

  void add(Node* &n, int tl, int tr, int l, int r, Line x){
    if(tr<=l||r<=tl) return;
    if(!n) n=new Node();
    if(l<=tl&&tr<=r) return addRec(n,tl,tr,x);
    int mid=(tl+tr)/2;
    add(n->lc,tl,mid,l,r,x);
    add(n->rc,mid,tr,l,r,x);
  }

  T query(Node* &n, int tl, int tr, int x){
```

```
    if(!n) return -INF;
    if(tl+1==tr) return n->line.get(x);
    T res=n->line.get(x);
    int mid=(tl+tr)/2;
    if(x<mid) res=max(res, query(n->lc,tl,mid,x));
    else res=max(res, query(n->rc,mid,tr,x));
    return res;
  }

  void add(Line x){add(root,0,sz,0,sz,x);}
  void add(Line x, int l, int r){add(root,0,sz,l,r,x);}
  T query(int x){return query(root,0,sz,x);}
};
```

2.17 Gain-cost-set

```
// stores pairs (benefit,cost) (erases non-optimal pairs)
struct GCS {
  set<pair<int,int> > s;
  void add(int g, int c){
    pair<int,int> x={g,c};
    auto p=s.lower_bound(x);
    if(p!=s.end()&&p->second<=x.second)return;
    if(p!=s.begin()){ // erase pairs with less benefit
      --p;            // and more cost
      while(p->second>=x.second){
        if(p==s.begin()){s.erase(p);break;}
        s.erase(p--);
      }
    }
    s.insert(x);
  }
  int get(int gain){ // min cost for some benefit
    auto p=s.lower_bound((pair<int,int>){gain,-INF});
    int r=p==s.end()?INF:p->second;
    return r;
  }
};
```

2.18 Disjoint intervals

```
// stores disjoint intervals as [first, second)
struct disjoint_intervals {
  set<pair<int,int> > s;
  void insert(pair<int,int> v){
    if(v.first>=v.second) return;
    auto at=s.lower_bound(v);auto it=at;
    if(at!=s.begin()&&(--at)->second>=v.first)v.first=at->first,--it;
    for(;it!=s.end()&&it->first<=v.second;s.erase(it++))
      v.second=max(v.second,it->second);
    segs.insert(v);
  }
};
```

3 Graphs

3.1 Topological sort

```
vector<int> g[MAXN];int n;
vector<int> tsort(){  // lexicographically smallest topological sort
  vector<int> r;priority_queue<int> q;
  vector<int> d(2*n,0);
  for (int i = 0; i < n; ++i)for (int j = 0; j < g[i].size(); ++j)d[g[i
      ][j]]++;
  for (int i = 0; i < n; ++i)if(!d[i])q.push(-i);
  while(!q.empty()){
    int x=-q.top();q.pop();r.push_back(x);
    for (int i = 0; i < g[x].size(); ++i){
      d[g[x][i]]--;
      if(!d[g[x][i]])q.push(-g[x][i]);
    }
  }
  return r;  // if not DAG it will have less than n elements
}
```

3.2 Kruskal (+ Union-Find)

```
int uf[MAXN];
void uf_init(){memset(uf,-1,sizeof(uf));}
int uf_find(int x){return uf[x]<0?x:uf[x]=uf_find(uf[x]);}
bool uf_join(int x, int y){
  x=uf_find(x);y=uf_find(y);
  if(x==y)return false;
  if(uf[x]>uf[y])swap(x,y);
  uf[x]+=uf[y];uf[y]=x;
  return true;
}
```

```cpp
vector<pair<ll,pair<int,int> > > es; // edges (cost,(u,v))
ll kruskal(){  // assumes graph is connected
  sort(es.begin(),es.end());uf_init();
  ll r=0;
  for (int i = 0; i < es.size(); ++i){
    int x=es[i].second.first,y=es[i].second.second;
    if(uf_join(x,y))r+=es[i].first; // (x,y,c) belongs to mst
  }
  return r; // total cost
}
```

3.3 Dijkstra

```cpp
vector<pair<int,int> > g[MAXN];  // u->[(v,cost)]
ll dist[MAXN];
void dijkstra(int x){
  memset(dist,-1,sizeof(dist));
  priority_queue<pair<ll,int> > q;
  dist[x]=0;q.push({0,x});
  while(!q.empty()){
    x=q.top().second;ll c=-q.top().first;q.pop();
    if(dist[x]!=c)continue;
    for (int i = 0; i < g[x].size(); ++i){
      int y=g[x][i].first; ll c=g[x][i].second;
      if(dist[y]<0||dist[x]+c<dist[y])
        dist[y]=dist[x]+c,q.push({-dist[y],y});
    }
  }
}
```

3.4 Bellman-Ford

```cpp
int n;
vector<pair<int,int> > g[MAXN]; // u->[(v,cost)]
ll dist[MAXN];
void bford(int src){ // O(nm)
  fill(dist,dist+n,INF);dist[src]=0;
  for (int _ = 0; _ < n; ++_)for (int x = 0; x < n; ++x)if(dist[x]!=INF)
      for(auto t:g[x]){
    dist[t.first]=min(dist[t.first],dist[x]+t.second);
  }
  for (int x = 0; x < n; ++x)if(dist[x]!=INF)for(auto t:g[x]){
    if(dist[t.first]>dist[x]+t.second){
      // neg cycle: all nodes reachable from t.first have -INF distance
      // to reconstruct neg cycle: save "prev" of each node, go up from
          t.first until repeating a node. this node and all nodes
          between the two occurences form a neg cycle
    }
  }
}
```

3.5 Floyd-Warshall

```cpp
// g[i][j]: weight of edge (i, j) or INF if there's no edge
// g[i][i]=0
ll g[MAXN][MAXN];int n;
void floyd(){ // O(n^3) . Replaces g with min distances
  for (int k = 0; k < n; ++k)for (int i = 0; i < n; ++i)if(g[i][k]<INF)
      for (int j = 0; j < n; ++j)if(g[k][j]<INF)
    g[i][j]=min(g[i][j],g[i][k]+g[k][j]);
}
bool inNegCycle(int v){return g[v][v]<0;}
bool hasNegCycle(int a, int b){ // true iff there's neg cycle in between
  for (int i = 0; i < n; ++i)if(g[a][i]<INF&&g[i][b]<INF&&g[i][i]<0)
      return true;
  return false;
}
```

3.6 Strongly connected components (+ 2-SAT)

```cpp
// MAXN: max number of nodes or 2 * max number of variables (2SAT)
bool truth[MAXN]; // truth[cmp[i]]=value of variable i (2SAT)
int nvar;int neg(int x){return MAXN-1-x;} // (2SAT)
vector<int> g[MAXN];
int n,lw[MAXN],idx[MAXN],qidx,cmp[MAXN],qcmp;
stack<int> st;
void tjn(int u){
  lw[u]=idx[u]=++qidx;
  st.push(u);cmp[u]=-2;
  for(int v:g[u]){
    if(!idx[v]||cmp[v]==-2){
      if(!idx[v]) tjn(v);
      lw[u]=min(lw[u],lw[v]);
    }
  }
  if(lw[u]==idx[u]){
    int x,l=-1;
    do{x=st.top();st.pop();cmp[x]=qcmp;if(min(x,neg(x))<nvar)l=x;}
```

```
    while(x!=u);
    if(l!=-1)truth[qcmp]=(cmp[neg(l)]<0); // (2SAT)
    qcmp++;
  }
}
void scc(){
  memset(idx,0,sizeof(idx));qidx=0;
  memset(cmp,-1,sizeof(cmp));qcmp=0;
  for (int i = 0; i < n; ++i)if(!idx[i])tjn(i);
}
// Only for 2SAT:
void addor(int a, int b){g[neg(a)].push_back(b);g[neg(b)].push_back(a);}
bool satisf(int _nvar){
  nvar=_nvar;n=MAXN;scc();
  for (int i = 0; i < nvar; ++i)if(cmp[i]==cmp[neg(i)])return false;
  return true;
}
```

3.7 Articulation - Bridges - Biconnected

```
vector<int> g[MAXN];int n;
struct edge {int u,v,comp;bool bridge;};
vector<edge> e;
void add_edge(int u, int v){
  g[u].push_back(e.size());g[v].push_back(e.size());
  e.push_back((edge){u,v,-1,false});
}
int D[MAXN],B[MAXN],T;
int nbc;  // number of biconnected components
int art[MAXN];  // articulation point iff !=0
stack<int> st;  // only for biconnected
void dfs(int u,int pe){
  B[u]=D[u]=T++;
  for(int ne:g[u])if(ne!=pe){
    int v=e[ne].u^e[ne].v^u;
    if(D[v]<0){
      st.push(ne);dfs(v,ne);
      if(B[v]>D[u])e[ne].bridge = true; // bridge
      if(B[v]>=D[u]){
        art[u]++; // articulation
        int last; // start biconnected
        do {
          last=st.top();st.pop();
          e[last].comp=nbc;
        } while(last!=ne);
        nbc++;    // end biconnected
      }
      B[u]=min(B[u],B[v]);
    }
    else if(D[v]<D[u])st.push(ne),B[u]=min(B[u],D[v]);
  }
}
void doit(){
  memset(D,-1,sizeof(D));memset(art,0,sizeof(art));
  nbc=T=0;
  for (int i = 0; i < n; ++i)if(D[i]<0)dfs(i,-1),art[i]--;
}
```

3.8 Chu-Liu (minimum spanning arborescence)

```
//O(n*m) minimum spanning tree in directed graph
//returns -1 if not possible
//included i-th edge if take[i]!=0
typedef int tw; tw INF=1ll<<30;
struct edge{int u,v,id;tw len;};
struct ChuLiu{
  int n; vector<edge> e;
  vector<int> inc,dec,take,pre,num,id,vis;
  vector<tw> inw;
  void add_edge(int x, int y, tw w){
    inc.push_back(0); dec.push_back(0); take.push_back(0);
    e.push_back({x,y,SZ(e),w});
  }
  ChuLiu(int n):n(n),pre(n),num(n),id(n),vis(n),inw(n){}
  tw doit(int root){
    auto e2=e;
    tw ans=0; int eg=SZ(e)-1,pos=SZ(e)-1;
    while(1){
      for (int i = 0; i < n; ++i) inw[i]=INF,id[i]=vis[i]=-1;
      for(auto ed:e2) if(ed.len<inw[ed.v]){
        inw[ed.v]=ed.len; pre[ed.v]=ed.u;
        num[ed.v]=ed.id;
      }
      inw[root]=0;
      for (int i = 0; i < n; ++i) if(inw[i]==INF) return -1;
      int tot=-1;
```

```
      for (int i = 0; i < n; ++i){
        ans+=inw[i];
        if(i!=root)take[num[i]]++;
        int j=i;
        while(vis[j]!=i&&j!=root&&id[j]<0)vis[j]=i,j=pre[j];
        if(j!=root&&id[j]<0){
          id[j]=++tot;
          for(int k=pre[j];k!=j;k=pre[k]) id[k]=tot;
        }
      }
      if(tot<0)break;
      for (int i = 0; i < n; ++i) if(id[i]<0)id[i]=++tot;
      n=tot+1; int j=0;
      for (int i = 0; i < SZ(e2); ++i){
        int v=e2[i].v;
        e2[j].v=id[e2[i].v];
        e2[j].u=id[e2[i].u];
        if(e2[j].v!=e2[j].u){
          e2[j].len=e2[i].len-inw[v];
          inc.push_back(e2[i].id);
          dec.push_back(num[v]);
          take.push_back(0);
          e2[j++].id=++pos;
        }
      }
      e2.resize(j);
      root=id[root];
    }
    while(pos>eg){
      if(take[pos]>0) take[inc[pos]]++, take[dec[pos]]--;
      pos--;
    }
    return ans;
  }
};
```

3.9 LCA - Binary Lifting

```
vector<int> g[1<<K];int n;  // K such that 2^K>=n
int F[K][1<<K],D[1<<K];
void lca_dfs(int x){
  for (int i = 0; i < g[x].size(); ++i){
    int y=g[x][i];if(y==F[0][x])continue;
    F[0][y]=x;D[y]=D[x]+1;lca_dfs(y);
  }
}
void lca_init(){
  D[0]=0;F[0][0]=-1;
  lca_dfs(0);
  for (int k = 1; k < K; ++k)for (int x = 0; x < n; ++x)
    if(F[k-1][x]<0)F[k][x]=-1;
    else F[k][x]=F[k-1][F[k-1][x]];
}
int lca(int x, int y){
  if(D[x]<D[y])swap(x,y);
  for(int k=K-1;k>=0;--k)if(D[x]-(1<<k)>=D[y])x=F[k][x];
  if(x==y)return x;
  for(int k=K-1;k>=0;--k)if(F[k][x]!=F[k][y])x=F[k][x],y=F[k][y];
  return F[0][x];
}
```

3.10 Heavy-Light decomposition

```
vector<int> g[MAXN];
int wg[MAXN],dad[MAXN],dep[MAXN]; // weight,father,depth
void dfs1(int x){
  wg[x]=1;
  for(int y:g[x])if(y!=dad[x]){
    dad[y]=x;dep[y]=dep[x]+1;dfs1(y);
    wg[x]+=wg[y];
  }
}
int curpos,pos[MAXN],head[MAXN];
void hld(int x, int c){
  if(c<0)c=x;
  pos[x]=curpos++;head[x]=c;
  int mx=-1;
  for(int y:g[x])if(y!=dad[x]&&(mx<0||wg[mx]<wg[y]))mx=y;
  if(mx>=0)hld(mx,c);
  for(int y:g[x])if(y!=mx&&y!=dad[x])hld(y,-1);
}
void hld_init(){dad[0]=-1;dep[0]=0;dfs1(0);curpos=0;hld(0,-1);}
int query(int x, int y, STree& rmq){
  int r=NEUT;
  while(head[x]!=head[y]){
    if(dep[head[x]]>dep[head[y]])swap(x,y);
```

```
    r=oper(r,rmq.query(pos[head[y]],pos[y]+1));
    y=dad[head[y]];
  }
  if(dep[x]>dep[y])swap(x,y); // now x is lca
  r=oper(r,rmq.query(pos[x],pos[y]+1));
  return r;
}
// for updating: rmq.upd(pos[x],v);
// queries on edges: - assign values of edges to "child" node
//                   - change pos[x] to pos[x]+1 in query (line 28)
```

3.11 Centroid decomposition

```
vector<int> g[MAXN];int n;
bool tk[MAXN];
int fat[MAXN]; // father in centroid decomposition
int szt[MAXN]; // size of subtree
int calcsz(int x, int f){
  szt[x]=1;
  for(auto y:g[x])if(y!=f&&!tk[y])szt[x]+=calcsz(y,x);
  return szt[x];
}
void cdfs(int x=0, int f=-1, int sz=-1){ // O(nlogn)
  if(sz<0)sz=calcsz(x,-1);
  for(auto y:g[x])if(!tk[y]&&szt[y]*2>=sz){
    szt[x]=0;cdfs(y,f,sz);return;
  }
  tk[x]=true;fat[x]=f;
  for(auto y:g[x])if(!tk[y])cdfs(y,x);
}
void centroid(){memset(tk,false,sizeof(tk));cdfs();}
```

3.12 Parallel DFS

```
struct Tree {
  int n,z[2];
  vector<vector<int>> g;
  vector<int> ex,ey,p,w,f,v[2];
  Tree(int n):g(n),w(n),f(n){}
  void add_edge(int x, int y){
    p.push_back(g[x].size());g[x].push_back(ex.size());ex.push_back(x);
        ey.push_back(y);
    p.push_back(g[y].size());g[y].push_back(ex.size());ex.push_back(y);
        ey.push_back(x);
  }
  bool go(int k){ // returns true if it finds new node
    int& x=z[k];
    while(x>=0&&
      (w[x]==g[x].size()||w[x]==g[x].size()-1&&(g[x].back()^1)==f[x]))
      x=f[x]>=0?ex[f[x]]:-1;
    if(x<0)return false;
    if((g[x][w[x]]^1)==f[x])w[x]++;
    int e=g[x][w[x]],y=ey[e];
    f[y]=e;w[x]++;w[y]=0;x=y;
    v[k].push_back(x);
    return true;
  }
  vector<int> erase_edge(int e){
    e*=2; // erases eth edge, returns smaller component
    int x=ex[e],y=ey[e];
    p[g[x].back()]=p[e];
    g[x][p[e]]=g[x].back();g[x].pop_back();
    p[g[y].back()]=p[e^1];
    g[y][p[e^1]]=g[y].back();g[y].pop_back();
    f[x]=f[y]=-1;
    w[x]=w[y]=0;
    z[0]=x;z[1]=y;
    v[0]={x};v[1]={y};
    bool d0=true,d1=true;
    while(d0&&d1)d0=go(0),d1=go(1);
    if(d1)return v[0];
    return v[1];
  }
};
```

3.13 Eulerian path

```
// Directed version (uncomment commented code for undirected)
struct edge {
  int y;
//  list<edge>::iterator rev;
  edge(int y):y(y){}
};
list<edge> g[MAXN];
void add_edge(int a, int b){
  g[a].push_front(edge(b));//auto ia=g[a].begin();
//  g[b].push_front(edge(a));auto ib=g[b].begin();
```

```
//  ia->rev=ib;ib->rev=ia;
}
vector<int> p;
void go(int x){
  while(g[x].size()){
    int y=g[x].front().y;
    //g[y].erase(g[x].front().rev);
    g[x].pop_front();
    go(y);
  }
  p.push_back(x);
}
vector<int> get_path(int x){ // get a path that begins in x
// check that a path exists from x before calling to get_path!
  p.clear();go(x);reverse(p.begin(),p.end());
  return p;
}
```

3.14 Dynamic connectivity

```
struct UnionFind {
  int n,comp;
  vector<int> uf,si,c;
  UnionFind(int n=0):n(n),comp(n),uf(n),si(n,1){
    for (int i = 0; i < n; ++i)uf[i]=i;}
  int find(int x){return x==uf[x]?x:find(uf[x]);}
  bool join(int x, int y){
    if((x=find(x))==(y=find(y)))return false;
    if(si[x]<si[y])swap(x,y);
    si[x]+=si[y];uf[y]=x;comp--;c.push_back(y);
    return true;
  }
  int snap(){return c.size();}
  void rollback(int snap){
    while(c.size()>snap){
      int x=c.back();c.pop_back();
      si[uf[x]]-=si[x];uf[x]=x;comp++;
    }
  }
};
enum {ADD,DEL,QUERY};
struct Query {int type,x,y;};
struct DynCon {
  vector<Query> q;
  UnionFind dsu;
  vector<int> mt;
  map<pair<int,int>,int> last;
  DynCon(int n):dsu(n){}
  void add(int x, int y){
    if(x>y)swap(x,y);
    q.push_back((Query){ADD,x,y});mt.push_back(-1);last[{x,y}]=q.size()
        -1;
  }
  void remove(int x, int y){
    if(x>y)swap(x,y);
    q.push_back((Query){DEL,x,y});
    int pr=last[{x,y}];mt[pr]=q.size()-1;mt.push_back(pr);
  }
  void query(){q.push_back((Query){QUERY,-1,-1});mt.push_back(-1);}
  void process(){ // answers all queries in order
    if(!q.size())return;
    for (int i = 0; i < q.size(); ++i)if(q[i].type==ADD&&mt[i]<0)mt[i]=q
        .size();
    go(0,q.size());
  }
  void go(int s, int e){
    if(s+1==e){
      if(q[s].type==QUERY) // answer query using DSU
        printf("%d\n",dsu.comp);
      return;
    }
    int k=dsu.snap(),m=(s+e)/2;
    for(int i=e-1;i>=m;--i)if(mt[i]>=0&&mt[i]<s)dsu.join(q[i].x,q[i].y);
    go(s,m);dsu.rollback(k);
    for(int i=m-1;i>=s;--i)if(mt[i]>=e)dsu.join(q[i].x,q[i].y);
    go(m,e);dsu.rollback(k);
  }
};
```

3.15 Dominator tree

```
//idom[i]=parent of i in dominator tree with root=rt, or -1 if not
    exists
int n,rnk[MAXN],pre[MAXN],anc[MAXN],idom[MAXN],semi[MAXN],low[MAXN];
vector<int> g[MAXN],rev[MAXN],dom[MAXN],ord;
void dfspre(int pos){
```

```
  rnk[pos]=SZ(ord); ord.push_back(pos);
  for(auto x:g[pos]){
    rev[x].push_back(pos);
    if(rnk[x]==n) pre[x]=pos,dfspre(x);
  }
}
int eval(int v){
  if(anc[v]<n&&anc[anc[v]]<n){
    int x=eval(anc[v]);
    if(rnk[semi[low[v]]]>rnk[semi[x]]) low[v]=x;
    anc[v]=anc[anc[v]];
  }
  return low[v];
}
void dominators(int rt){
  for (int i = 0; i < n; ++i){
    dom[i].clear(); rev[i].clear();
    rnk[i]=pre[i]=anc[i]=idom[i]=n;
    semi[i]=low[i]=i;
  }
  ord.clear(); dfspre(rt);
  for(int i=SZ(ord)-1;i;i--){
    int w=ord[i];
    for(int v:rev[w]){
      int u=eval(v);
      if(rnk[semi[w]]>rnk[semi[u]])semi[w]=semi[u];
    }
    dom[semi[w]].push_back(w); anc[w]=pre[w];
    for(int v:dom[pre[w]]){
      int u=eval(v);
      idom[v]=(rnk[pre[w]]>rnk[semi[u]]?u:pre[w]);
    }
    dom[pre[w]].clear();
  }
  for(int w:ord) if(w!=rt&&idom[w]!=semi[w]) idom[w]=idom[idom[w]];
  for (int i = 0; i < n; ++i) if(idom[i]==n)idom[i]=-1;
}
```

3.16 Edmond's blossom (matching in general graphs)

```
vector<int> g[MAXN];
int n,m,mt[MAXN],qh,qt,q[MAXN],ft[MAXN],bs[MAXN];
bool inq[MAXN],inb[MAXN],inp[MAXN];
int lca(int root, int x, int y){
  memset(inp,0,sizeof(inp));
  while(1){
    inp[x=bs[x]]=true;
    if(x==root)break;
    x=ft[mt[x]];
  }
  while(1){
    if(inp[y=bs[y]])return y;
    else y=ft[mt[y]];
  }
}
void mark(int z, int x){
  while(bs[x]!=z){
    int y=mt[x];
    inb[bs[x]]=inb[bs[y]]=true;
    x=ft[y];
    if(bs[x]!=z)ft[x]=y;
  }
}
void contr(int s, int x, int y){
  int z=lca(s,x,y);
  memset(inb,0,sizeof(inb));
  mark(z,x);mark(z,y);
  if(bs[x]!=z)ft[x]=y;
  if(bs[y]!=z)ft[y]=x;
  for (int x = 0; x < n; ++x)if(inb[bs[x]]){
    bs[x]=z;
    if(!inq[x])inq[q[++qt]=x]=true;
  }
}
int findp(int s){
  memset(inq,0,sizeof(inq));
  memset(ft,-1,sizeof(ft));
  for (int i = 0; i < n; ++i)bs[i]=i;
  inq[q[qh=qt=0]=s]=true;
  while(qh<=qt){
    int x=q[qh++];
    for(int y:g[x])if(bs[x]!=bs[y]&&mt[x]!=y){
      if(y==s||mt[y]>=0&&ft[mt[y]]>=0)contr(s,x,y);
      else if(ft[y]<0){
        ft[y]=x;
        if(mt[y]<0)return y;
```

```
        else if(!inq[mt[y]])inq[q[++qt]=mt[y]]=true;
      }
    }
  }
  return -1;
}
int aug(int s, int t){
  int x=t,y,z;
  while(x>=0){
    y=ft[x];
    z=mt[y];
    mt[y]=x;mt[x]=y;
    x=z;
  }
  return t>=0;
}
int edmonds(){ // O(n^2 m)
  int r=0;
  memset(mt,-1,sizeof(mt));
  for (int x = 0; x < n; ++x)if(mt[x]<0)r+=aug(x,findp(x));
  return r;
}
```

4 Math

4.1 Identities

$$C_n = \frac{2(2n-1)}{n+1}C_{n-1}$$
$$C_n = \frac{1}{n+1}\binom{2n}{n}$$
$$C_n \sim \frac{4^n}{n^{3/2}\sqrt{\pi}}$$
$$\sigma(n) = O(\log(\log(n))) \text{ (number of divisors of } n)$$
$$F_{2n+1} = F_n^2 + F_{n+1}^2$$
$$F_{2n} = F_{n+1}^2 - F_{n-1}^2$$
$$\sum_{i=1}^{n} F_i = F_{n+2} - 1$$
$$F_{n+i}F_{n+j} - F_nF_{n+i+j} = (-1)^n F_iF_j$$
(Möbius Inv. Formula) Let $g(n) = \sum_{d|n} f(d)$, then $f(n) = \sum d \mid ng(d)\mu\left(\frac{n}{d}\right)$.

4.2 Theorems

```
(Tutte) A graph, G = (V, E), has a perfect matching if and only if for
    every subset U of V, the subgraph induced by V - U has at most |U|
    connected components with an odd number of vertices.
Petersens Theorem. Every cubic, bridgeless graph contains a perfect
    matching.
(Dilworth) In any finite partially ordered set, the maximum number of
    elements in any antichain equals the minimum number of chains in any
     partition of the set into chains
Pick: A=I+B/2-1  (area of polygon, points inside, points on border)
```

4.3 Integer floor division

```
void floordiv(ll x, ll y, ll& q, ll& r) { // (for negative x)
  q=x/y;r=x%y;
  if((r!=0)&&((r<0)!=(y<0)))q--,r+=y;
}
```

4.4 Sieve of Eratosthenes

```
int cr[MAXN]; // -1 if prime, some not trivial divisor if not
void init_sieve(){
  memset(cr,-1,sizeof(cr));
  for (int i = 2; i < MAXN; ++i)if(cr[i]<0)for(ll j=1LL*i*i;j<MAXN;j+=i)
      cr[j]=i;
}
map<int,int> fact(int n){  // must call init_cribe before
  map<int,int> r;
  while(cr[n]>=0)r[cr[n]]++,n/=cr[n];
  if(n>1)r[n]++;
  return r;
}
```

4.5 Generate divisors

```
void div_rec(vector<ll>& r, vector<pair<ll,int> >& f, int k, ll c){
  if(k==f.size()){r.push_back(c);return;}
  for (int i = 0; i < f[k].second+1; ++i)div_rec(r,f,k+1,c),c*=f[k].
      first;
}
vector<ll> divisors(vector<pair<ll,int> > f){
  vector<ll> r; // returns divisors given factorization
  div_rec(r,f,0,1);
  return r;
}
```

4.6 Pollard's rho

```cpp
ll gcd(ll a, ll b){return a?gcd(b%a,a):b;}
ll mulmod(ll a, ll b, ll m) {
  ll r=a*b-(ll)((long double)a*b/m+.5)*m;
  return r<0?r+m:r;
}
ll expmod(ll b, ll e, ll m){
  if(!e)return 1;
  ll q=expmod(b,e/2,m);q=mulmod(q,q,m);
  return e&1?mulmod(b,q,m):q;
}
bool is_prime_prob(ll n, int a){
  if(n==a)return true;
  ll s=0,d=n-1;
  while(d%2==0)s++,d/=2;
  ll x=expmod(a,d,n);
  if((x==1)||(x+1==n))return true;
  for (int _ = 0; _ < s-1; ++_){
    x=mulmod(x,x,n);
    if(x==1)return false;
    if(x+1==n)return true;
  }
  return false;
}
bool rabin(ll n){ // true iff n is prime
  if(n==1)return false;
  int ar[]={2,3,5,7,11,13,17,19,23};
  for (int i = 0; i < 9; ++i)if(!is_prime_prob(n,ar[i]))return false;
  return true;
}
ll rho(ll n){
  if(!(n&1))return 2;
  ll x=2,y=2,d=1;
  ll c=rand()%n+1;
  while(d==1){
    x=(mulmod(x,x,n)+c)%n;
    y=(mulmod(y,y,n)+c)%n;
    y=(mulmod(y,y,n)+c)%n;
    if(x>=y)d=gcd(x-y,n);
    else d=gcd(y-x,n);
  }
  return d==n?rho(n):d;
}
void fact(ll n, map<ll,int>& f){ //O (lg n)^3
  if(n==1)return;
  if(rabin(n)){f[n]++;return;}
  ll q=rho(n);fact(q,f);fact(n/q,f);
}
// optimized version: replace rho and fact with the following:
const int MAXP=1e6+1; // sieve size
int sv[MAXP]; // sieve
ll add(ll a, ll b, ll m){return (a+=b)<m?a:a-m;}
ll rho(ll n){
  static ll s[MAXP];
  while(1){
    ll x=rand()%n,y=x,c=rand()%n;
    ll *px=s,*py=s,v=0,p=1;
    while(1){
      *py++=y=add(mulmod(y,y,n),c,n);
      *py++=y=add(mulmod(y,y,n),c,n);
      if((x=*px++)==y)break;
      ll t=p;
      p=mulmod(p,abs(y-x),n);
      if(!p)return gcd(t,n);
      if(++v==26){
        if((p=gcd(p,n))>1&&p<n)return p;
        v=0;
      }
    }
    if(v&&(p=gcd(p,n))>1&&p<n)return p;
  }
}
void init_sv(){
  for (int i = 2; i < MAXP; ++i)if(!sv[i])for(ll j=i;j<MAXP;j+=i)sv[j]=i
      ;
}
void fact(ll n, map<ll,int>& f){ // call init_sv first!!!
  for(auto&& p:f){
    while(n%p.first==0){
      p.second++;
      n/=p.first;
    }
  }
  if(n<MAXP)while(n>1)f[sv[n]]++,n/=sv[n];
  else if(rabin(n))f[n]++;
  else {ll q=rho(n);fact(q,f);fact(n/q,f);}
}
```

4.7 Simpson's rule

```
double integrate(double f(double), double a, double b, int n=10000){
  double r=0,h=(b-a)/n,fa=f(a),fb;
  for (int i = 0; i < n; ++i){fb=f(a+h*(i+1));r+=fa+4*f(a+h*(i+0.5))+fb;
      fa=fb;}
  return r*h/6.;
}
```

4.8 Polynomials

```
typedef int tp; // type of polynomial
template<class T=tp>
struct poly {  // poly<> : 1 variable, poly<poly<>>: 2 variables, etc.
  vector<T> c;
  T& operator[](int k){return c[k];}
  poly(vector<T>& c):c(c){}
  poly(initializer_list<T> c):c(c){}
  poly(int k):c(k){}
  poly(){}
  poly operator+(poly<T> o){
    int m=c.size(),n=o.c.size();
    poly res(max(m,n));
    for (int i = 0; i < m; ++i)res[i]=res[i]+c[i];
    for (int i = 0; i < n; ++i)res[i]=res[i]+o.c[i];
    return res;
  }
  poly operator*(tp k){
    poly res(c.size());
    for (int i = 0; i < c.size(); ++i)res[i]=c[i]*k;
    return res;
  }
  poly operator*(poly o){
    int m=c.size(),n=o.c.size();
    poly res(m+n-1);
    for (int i = 0; i < m; ++i)for (int j = 0; j < n; ++j)res[i+j]=res[i
        +j]+c[i]*o.c[j];
    return res;
  }
  poly operator-(poly<T> o){return *this+(o*-1);}
  T operator()(tp v){
    T sum(0);
    for(int i=c.size()-1;i>=0;--i)sum=sum*v+c[i];
    return sum;
  }
};
// example: p(x,y)=2*x^2+3*x*y-y+4
// poly<poly<>> p={{4,-1},{0,3},{2}}
// printf("%d\n",p(2)(3)) // 27 (p(2,3))
set<tp> roots(poly<> p){ // only for integer polynomials
  set<tp> r;
  while(!p.c.empty()&&!p.c.back())p.c.pop_back();
  if(!p(0))r.insert(0);
  if(p.c.empty())return r;
  tp a0=0,an=abs(p[p.c.size()-1]);
  for(int k=0;!a0;a0=abs(p[k++]));
  vector<tp> ps,qs;
  for (int i = 1; i < sqrt(a0)+1; ++i)if(a0%i==0)ps.push_back(i),ps.
      push_back(a0/i);
  for (int i = 1; i < sqrt(an)+1; ++i)if(an%i==0)qs.push_back(i),qs.
      push_back(an/i);
  for(auto pt:ps)for(auto qt:qs)if(pt%qt==0){
    tp x=pt/qt;
    if(!p(x))r.insert(x);
    if(!p(-x))r.insert(-x);
  }
  return r;
}
pair<poly<>,tp> ruffini(poly<> p, tp r){ // returns pair (result,rem)
  int n=p.c.size()-1;
  vector<tp> b(n);
  b[n-1]=p[n];
  for(int k=n-2;k>=0;--k)b[k]=p[k+1]+r*b[k+1];
  return {poly<>(b),p[0]+r*b[0]};
}
// only for double polynomials
pair<poly<>,poly<> > polydiv(poly<> p, poly<> q){ // returns pair (
    result,rem)
  int n=p.c.size()-q.c.size()+1;
  vector<tp> b(n);
  for(int k=n-1;k>=0;--k){
    b[k]=p.c.back()/q.c.back();
    for (int i = 0; i < q.c.size(); ++i)p[i+k]-=b[k]*q[i];
    p.c.pop_back();
  }
  while(!p.c.empty()&&abs(p.c.back())<EPS)p.c.pop_back();
```

```
  return {poly<>(b),p};
}
// only for double polynomials
poly<> interpolate(vector<tp> x, vector<tp> y){ //TODO TEST
  poly<> q={1},S={0};
  for(tp a:x)q=poly<>({-a,1})*q;
  for (int i = 0; i < x.size(); ++i){
    poly<> Li=ruffini(q,x[i]).first;
    Li=Li*(1.0/Li(x[i])); // change for int polynomials
    S=S+Li*y[i];
  }
  return S;
}
```

4.9 Bairstow

```
double pget(poly<>& p, int k){return k<p.c.size()?p[k]:0;}
poly<> bairstow(poly<> p){ // returns polynomial of degree 2 that
  int n=p.c.size()-1;    // divides p
  assert(n>=3&&abs(p.c.back())>EPS);
  double u=p[n-1]/p[n],v=p[n-2]/p[n];
  for (int _ = 0; _ < ITER; ++_){
    auto w=polydiv(p,{v,u,1});
    poly<> q=w.first,r0=w.second;
    poly<> r1=polydiv(q,{v,u,1}).second;
    double c=pget(r0,1),d=pget(r0,0),g=pget(r1,1),h=pget(r1,0);
    double det=1/(v*g*g+h*(h-u*g)),uu=u;
    u-=det*(-h*c+g*d);v-=det*(-g*v*c+(g*uu-h)*d);

  }
  return {v,u,1};
}
void addr(vector<double>& r, poly<>& p){
  assert(p.c.size()<=3);
  if(p.c.size()<=1)return;
  if(p.c.size()==2)r.push_back(-p[0]/p[1]);
  if(p.c.size()==3){
    double a=p[2],b=p[1],c=p[0];
    double d=b*b-4*a*c;
    if(d<-0.1)return; // huge epsilon because of bad precision
    d=d>0?sqrt(d):0;r.push_back((-b-d)/2/a);r.push_back((-b+d)/2/a);
  }
}
```

```
vector<double> roots(poly<> p){
  while(!p.c.empty()&&abs(p.c.back())<EPS)p.c.pop_back();
  for (int i = 0; i < p.c.size(); ++i)p[i]/=p.c.back();
  vector<double> r;int n;
  while((n=p.c.size()-1)>=3){
    poly<> q=bairstow(p);addr(r,q);
    p=polydiv(p,q).first;
    while(p.c.size()>n-1)p.c.pop_back();
  }
  addr(r,p);
  return r;
}
```

4.10 Fast Fourier Transform

```
// MAXN must be power of 2 !!
// MOD-1 needs to be a multiple of MAXN !!
// big mod and primitive root for NTT:
typedef ll tf;
typedef vector<tf> poly;
const tf MOD=2305843009255636993,RT=5;
// FFT
struct CD {
  double r,i;
  CD(double r=0, double i=0):r(r),i(i){}
  double real()const{return r;}
  void operator/=(const int c){r/=c, i/=c;}
};
CD operator*(const CD& a, const CD& b){
  return CD(a.r*b.r-a.i*b.i,a.r*b.i+a.i*b.r);}
CD operator+(const CD& a, const CD& b){return CD(a.r+b.r,a.i+b.i);}
CD operator-(const CD& a, const CD& b){return CD(a.r-b.r,a.i-b.i);}
const double pi=acos(-1.0);
// NTT
/*
struct CD {
  tf x;
  CD(tf x):x(x){}
  CD(){}
};
CD operator*(const CD& a, const CD& b){return CD(mulmod(a.x,b.x));}
CD operator+(const CD& a, const CD& b){return CD(addmod(a.x,b.x));}
CD operator-(const CD& a, const CD& b){return CD(submod(a.x,b.x));}
```

```
vector<tf> rts(MAXN+9,-1);
CD root(int n, bool inv){
  tf r=rts[n]<0?rts[n]=pm(RT,(MOD-1)/n):rts[n];
  return CD(inv?pm(r,MOD-2):r);
}
*/
CD cp1[MAXN+9],cp2[MAXN+9];
int R[MAXN+9];
void dft(CD* a, int n, bool inv){
  for (int i = 0; i < n; ++i)if(R[i]<i)swap(a[R[i]],a[i]);
  for(int m=2;m<=n;m*=2){
    double z=2*pi/m*(inv?-1:1); // FFT
    CD wi=CD(cos(z),sin(z)); // FFT
    // CD wi=root(m,inv); // NTT
    for(int j=0;j<n;j+=m){
      CD w(1);
      for(int k=j,k2=j+m/2;k2<j+m;k++,k2++){
        CD u=a[k];CD v=a[k2]*w;a[k]=u+v;a[k2]=u-v;w=w*wi;
      }
    }
  }
  if(inv)for (int i = 0; i < n; ++i)a[i]/=n; // FFT
  //if(inv){ // NTT
  //  CD z(pm(n,MOD-2)); // pm: modular exponentiation
  //  for (int i = 0; i < n; ++i)a[i]=a[i]*z;
  //}
}
poly multiply(poly& p1, poly& p2){
  int n=p1.size()+p2.size()+1;
  int m=1,cnt=0;
  while(m<=n)m+=m,cnt++;
  for (int i = 0; i < m; ++i){R[i]=0;for (int j = 0; j < cnt; ++j)R[i]=(
      R[i]<<1)|((i>>j)&1);}
  for (int i = 0; i < m; ++i)cp1[i]=0,cp2[i]=0;
  for (int i = 0; i < p1.size(); ++i)cp1[i]=p1[i];
  for (int i = 0; i < p2.size(); ++i)cp2[i]=p2[i];
  dft(cp1,m,false);dft(cp2,m,false);
  for (int i = 0; i < m; ++i)cp1[i]=cp1[i]*cp2[i];
  dft(cp1,m,true);
  poly res;
  n-=2;
  for (int i = 0; i < n; ++i)res.push_back((tf)floor(cp1[i].real()+0.5))
      ; // FFT
  //for (int i = 0; i < n; ++i)res.push_back(cp1[i].x); // NTT
  return res;
}
```

4.11 FFT operations

```
// Polynomial division: O(n*log(n))
// Multi-point polynomial evaluation for arbitrarily large points: O(n*
    log^2(n))
// Multi-point polynomial evaluation for all points in [0, MOD): O((n+
    MOD)*log(n+MOD))
// Polynomial interpolation: O(n*log^2(n))
// Inverse: O(n*log(n))
// Log: O(n*log(n))
// Exp: O(n*log(n))

//Works with NTT. For FFT, just replace addmod,submod,mulmod,inv
poly add(poly &a, poly &b){
  int n=SZ(a),m=SZ(b);
  poly ans(max(n,m));
  for (int i = 0; i < max(n,m); ++i) ans[i]=addmod(i<SZ(a)?a[i]:0, i<SZ(
      b)?b[i]:0);
  while(SZ(ans)>1&&!ans.back())ans.pop_back();
  return ans;
}

// derivative of p
poly derivate(poly &p){
  poly ans(max(1, SZ(p)-1));
  for (int i = 1; i < SZ(p); ++i) ans[i-1]=mulmod(p[i],i);
  return ans;
}

// integral of p
poly integrate(poly &p){
  poly ans(SZ(p)+1);
  for (int i = 0; i < SZ(p); ++i) ans[i+1]=mulmod(p[i], inv(i+1));
  return ans;
}

// p % (x^n)
poly takemod(poly &p, int n){
  poly res=p;
```

```
  res.resize(min(SZ(res),n));
  while(SZ(res)>1&&res.back()==0) res.pop_back();
  return res;
}

// first d terms of 1/p
poly invert(poly &p, int d){
  assert(p[0]);
  poly res={inv(p[0])};
  int sz=1;
  while(sz<d){
    sz*=2;
    poly pre(p.begin(), p.begin()+min(SZ(p),sz));
    poly cur=multiply(res,pre);
    for (int i = 0; i < SZ(cur); ++i) cur[i]=submod(0,cur[i]);
    cur[0]=addmod(cur[0],2);
    res=multiply(res,cur);
    res=takemod(res,sz);
  }
  res.resize(d);
  return res;
}

// first d terms of log(p)
poly log(poly &p, int d){
  assert(p[0]==1);
  poly cur=takemod(p,d);
  poly a=invert(cur,d), b=derivate(cur);
  auto res=multiply(a,b);
  res=takemod(res,d-1);
  res=integrate(res);
  return res;
}

// first d terms of exp(p)
poly exp(poly &p, int d){
  assert(!p[0]);
  poly res={1};
  int sz=1;
  while(sz<d){
    sz*=2;
    poly lg=log(res, sz), cur(sz);
    for (int i = 0; i < sz; ++i) cur[i]=submod(i<SZ(p)?p[i]:0, i<SZ(lg)?
        lg[i]:0);
    cur[0]=addmod(cur[0],1);
    res=multiply(res,cur);
    res=takemod(res, sz);
  }

  res.resize(d);
  return res;
}

pair<poly,poly> divslow(poly &a, poly &b){
  poly q,r=a;
  while(SZ(r)>=SZ(b)){
    q.push_back(mulmod(r.back(),inv(b.back())));
    if(q.back()) for (int i = 0; i < SZ(b); ++i){
      r[SZ(r)-i-1]=submod(r[SZ(r)-i-1],mulmod(q.back(),b[SZ(b)-i-1]));
    }
    r.pop_back();
  }
  reverse(ALL(q));
  return {q,r};
}

pair<poly,poly> divide(poly &a, poly &b){ //returns {quotient,remainder}
  int m=SZ(a),n=SZ(b),MAGIC=750;
  if(m<n) return {{0},a};
  if(min(m-n,n)<MAGIC)return divslow(a,b);
  poly ap=a; reverse(ALL(ap));
  poly bp=b; reverse(ALL(bp));
  bp=invert(bp,m-n+1);
  poly q=multiply(ap,bp);
  q.resize(SZ(q)+m-n-SZ(q)+1,0);
  reverse(ALL(q));
  poly bq=multiply(b,q);
  for (int i = 0; i < SZ(bq); ++i) bq[i]=submod(0,bq[i]);
  poly r=add(a,bq);
  return {q,r};
}

vector<poly> tree;

void filltree(vector<tf> &x){
  int k=SZ(x);
```

```
  tree.resize(2*k);
  for (int i = k; i < 2*k; ++i) tree[i]={submod(0,x[i-k]),1};
  for(int i=k-1;i;i--) tree[i]=multiply(tree[2*i],tree[2*i+1]);
}

// Multi-point polynomial evaluation for arbitrarily large points
vector<tf> evaluate(poly &a, vector<tf> &x){
  filltree(x);
  int k=SZ(x);
  vector<poly> ans(2*k);
  ans[1]=divide(a,tree[1]).second;
  for (int i = 2; i < 2*k; ++i) ans[i]=divide(ans[i>>1],tree[i]).second;
  vector<tf> r; for (int i = 0; i < k; ++i) r.push_back(ans[i+k][0]);
  return r;
}

// Given any number g, perform multi-point polynomial evaluation
// for all points of the form g^i, for each i in [0, k)
// in O((n+k) * log(n+k))
vector<int> chirpTransform(poly& p, int g, int k) {
  int inv_g=inv(g), n=SZ(p)-1, sz=min(k,MOD-1), gk=1;
  poly ap(n+1), bp(n+sz);

  for (int i = 0; i < n+sz; ++i){
    ll exp=1ll*i*(i-1)/2%(MOD-1);
    if(i<=n) ap[n-i]=mulmod(p[i],fpow(inv_g, exp));
    bp[i]=fpow(g, exp);
  }

  poly cp=multiply(ap, bp);
  vector<int> ans(k);

  for (int i = 0; i < sz; ++i){
    ll exp=1ll*i*(i-1)/2%(MOD-1);
    int val=0;
    if(n+i<SZ(cp)) val=cp[n+i];
    val=mulmod(val, fpow(inv_g,exp));
    ans[i]=val;
    gk=mulmod(gk,g);
  }
  for (int i = sz; i < k; ++i) ans[i]=ans[i-MOD+1];
  return ans;
}

int get_primitive_root(int p) {
  if(p==2)return 1;
  int phi=p-1, n=phi;
  vector<int> fact;
  for(int i=2;i*i<=n;i++) if(n%i==0){
    fact.push_back(i);
    while(n%i==0) n/=i;
  }
  if(n>1) fact.push_back(n);

  for (int res = 2; res < p+1; ++res){
    int ok=1;
    for(int f:fact){
      if(fpow(res,phi/f)==1){
        ok=false;
        break;
      }
    }
    if(ok)return res;
  }
  return -1;
}

// Multi-point polynomial evaluation for all points in [0, MOD)
// MOD needs to be a prime number
vector<int> evaluate_all_points(poly& p) {
  int g=get_primitive_root(MOD), gk=1;
  assert(g!=-1);

  vector<int> ch=chirpTransform(p,g,MOD-1), ans(MOD, p[0]);

  for (int i = 0; i < MOD-1; ++i){
    ans[gk]=ch[i];
    gk=mulmod(gk,g);
  }
  return ans;
}

poly interpolate(vector<tf> &x, vector<tf> &y){
  filltree(x);
  poly p=derivate(tree[1]);
  int k=SZ(y);
```

```
  vector<tf> d=evaluate(p,x);
  vector<poly> intree(2*k);
  for (int i = k; i < 2*k; ++i) intree[i]={mulmod(y[i-k],inv(d[i-k]))};
  for(int i=k-1;i;i--){
    poly p1=multiply(tree[2*i],intree[2*i+1]);
    poly p2=multiply(tree[2*i+1],intree[2*i]);
    intree[i]=add(p1,p2);
  }
  return intree[1];
}
```

4.12 Fast Hadamard Transform

```
ll c1[MAXN+9],c2[MAXN+9];  // MAXN must be power of 2 !!
void fht(ll* p, int n, bool inv){
  for(int l=1;2*l<=n;l*=2)for(int i=0;i<n;i+=2*l)for (int j = 0; j < l;
      ++j){
    ll u=p[i+j],v=p[i+l+j];
    if(!inv)p[i+j]=u+v,p[i+l+j]=u-v; // XOR
    else p[i+j]=(u+v)/2,p[i+l+j]=(u-v)/2;
    //if(!inv)p[i+j]=v,p[i+l+j]=u+v; // AND
    //else p[i+j]=-u+v,p[i+l+j]=u;
    //if(!inv)p[i+j]=u+v,p[i+l+j]=u; // OR
    //else p[i+j]=v,p[i+l+j]=u-v;
  }
}
// like polynomial multiplication, but XORing exponents
// instead of adding them (also ANDing, ORing)
vector<ll> multiply(vector<ll>& p1, vector<ll>& p2){
  int n=1<<(32-__builtin_clz(max(SZ(p1),SZ(p2))-1));
  for (int i = 0; i < n; ++i)c1[i]=0,c2[i]=0;
  for (int i = 0; i < SZ(p1); ++i)c1[i]=p1[i];
  for (int i = 0; i < SZ(p2); ++i)c2[i]=p2[i];
  fht(c1,n,false);fht(c2,n,false);
  for (int i = 0; i < n; ++i)c1[i]*=c2[i];
  fht(c1,n,true);
  return vector<ll>(c1,c1+n);
}
```

4.13 Karatsuba

```
typedef ll tp;
#define add(n,s,d,k) for (int i = 0; i < n; ++i)(d)[i]+=(s)[i]*k
tp* ini(int n){tp *r=new tp[n];fill(r,r+n,0);return r;}
void karatsura(int n, tp* p, tp* q, tp* r){
  if(n<=0)return;
  if(n<35)for (int i = 0; i < n; ++i)for (int j = 0; j < n; ++j)r[i+j]+=
      p[i]*q[j];
  else {
    int nac=n/2,nbd=n-n/2;
    tp *a=p,*b=p+nac,*c=q,*d=q+nac;
    tp *ab=ini(nbd+1),*cd=ini(nbd+1),*ac=ini(nac*2),*bd=ini(nbd*2);
    add(nac,a,ab,1);add(nbd,b,ab,1);
    add(nac,c,cd,1);add(nbd,d,cd,1);
    karatsura(nac,a,c,ac);karatsura(nbd,b,d,bd);
    add(nac*2,ac,r+nac,-1);add(nbd*2,bd,r+nac,-1);
    add(nac*2,ac,r,1);add(nbd*2,bd,r+nac*2,1);
    karatsura(nbd+1,ab,cd,r+nac);
    free(ab);free(cd);free(ac);free(bd);
  }
}
vector<tp> multiply(vector<tp> p0, vector<tp> p1){
  int n=max(p0.size(),p1.size());
  tp *p=ini(n),*q=ini(n),*r=ini(2*n);
  for (int i = 0; i < p0.size(); ++i)p[i]=p0[i];
  for (int i = 0; i < p1.size(); ++i)q[i]=p1[i];
  karatsura(n,p,q,r);
  vector<tp> rr(r,r+p0.size()+p1.size()-1);
  free(p);free(q);free(r);
  return rr;
}
```

4.14 Diophantine

```
pair<ll,ll> extendedEuclid (ll a, ll b){ //a * x + b * y = gcd(a,b)
  ll x,y;
  if (b==0) return {1,0};
  auto p=extendedEuclid(b,a%b);
  x=p.second;
  y=p.first-(a/b)*x;
  if(a*x+b*y==-gcd(a,b)) x=-x, y=-y;
  return {x,y};
}
pair<pair<ll,ll>,pair<ll,ll> > diophantine(ll a,ll b, ll r) {
  //a*x+b*y=r where r is multiple of gcd(a,b);
  ll d=gcd(a,b);
  a/=d; b/=d; r/=d;
```

```
  auto p = extendedEuclid(a,b);
  p.first*=r; p.second*=r;
  assert(a*p.first+b*p.second==r);
  return {p,{-b,a}}; // solutions: p+t*ans.second
}
```

4.15 Modular inverse

```
ll inv(ll a, ll mod) { //inverse of a modulo mod
  assert(gcd(a,mod)==1);
  pl sol = extendedEuclid(a,mod);
  return ((sol.first%mod)+mod)%mod;
}
```

4.16 Chinese remainder theorem

```
#define mod(a,m) (((a)%m+m)%m)
pair<ll,ll> sol(tuple<ll,ll,ll> c){ //requires inv, diophantine
    ll a=get<0>(c), x1=get<1>(c), m=get<2>(c), d=gcd(a,m);
    if(d==1) return {mod(x1*inv(a,m),m), m};
    else return x1%d ? ii({-1LL,-1LL}) : sol(make_tuple(a/d,x1/d,m/d));
}
pair<ll,ll> crt(vector< tuple<ll,ll,ll> > cond) { // returns: (sol, lcm)
  ll x1=0,m1=1,x2,m2;
  for(auto t:cond){
    tie(x2,m2)=sol(t);
    if(m2<0||(x1-x2)%gcd(m1,m2))return {-1,-1};
    if(m1==m2)continue;
    ll k=diophantine(m2,-m1,x1-x2).first.second,l=m1*(m2/gcd(m1,m2));
    x1=mod((__int128)m1*k+x1,l);m1=l;
  }
  return sol(make_tuple(1,x1,m1));
} //cond[i]={ai,bi,mi} ai*xi=bi (mi); assumes lcm fits in ll
```

4.17 Mobius

```
short mu[MAXN] = {0,1};
void mobius(){
  for (int i = 1; i < MAXN; ++i)if(mu[i])for(int j=i+i;j<MAXN;j+=i)mu[j
      ]-=mu[i];
}
```

4.18 Matrix exponentiation

```
typedef vector<vector<ll> > Matrix;
Matrix ones(int n) {
  Matrix r(n,vector<ll>(n));
  for (int i = 0; i < n; ++i)r[i][i]=1;
  return r;
}
Matrix operator*(Matrix &a, Matrix &b) {
  int n=SZ(a),m=SZ(b[0]),z=SZ(a[0]);
  Matrix r(n,vector<ll>(m));
  for (int i = 0; i < n; ++i)for (int j = 0; j < m; ++j)for (int k = 0;
      k < z; ++k)
    r[i][j]+=a[i][k]*b[k][j],r[i][j]%=mod;
  return r;
}
Matrix be(Matrix b, ll e) {
  Matrix r=ones(SZ(b));
  while(e){if(e&1LL)r=r*b;b=b*b;e/=2;}
  return r;
}
```

4.19 Matrix reduce and determinant

```
double reduce(vector<vector<double> >& x){ // returns determinant
  int n=x.size(),m=x[0].size();
  int i=0,j=0;double r=1.;
  while(i<n&&j<m){
    int l=i;
    for (int k = i+1; k < n; ++k)if(abs(x[k][j])>abs(x[l][j]))l=k;
    if(abs(x[l][j])<EPS){j++;r=0.;continue;}
    if(l!=i){r=-r;swap(x[i],x[l]);}
    r*=x[i][j];
    for(int k=m-1;k>=j;k--)x[i][k]/=x[i][j];
    for (int k = 0; k < n; ++k){
      if(k==i)continue;
      for(int l=m-1;l>=j;l--)x[k][l]-=x[k][j]*x[i][l];
    }
    i++;j++;
  }
  return r;
}
```

4.20 Simplex

```
vector<int> X,Y;
```

```
vector<vector<double> > A;
vector<double> b,c;
double z;
int n,m;
void pivot(int x,int y){
  swap(X[y],Y[x]);
  b[x]/=A[x][y];
  for (int i = 0; i < m; ++i)if(i!=y)A[x][i]/=A[x][y];
  A[x][y]=1/A[x][y];
  for (int i = 0; i < n; ++i)if(i!=x&&abs(A[i][y])>EPS){
    b[i]-=A[i][y]*b[x];
    for (int j = 0; j < m; ++j)if(j!=y)A[i][j]-=A[i][y]*A[x][j];
    A[i][y]=-A[i][y]*A[x][y];
  }
  z+=c[y]*b[x];
  for (int i = 0; i < m; ++i)if(i!=y)c[i]-=c[y]*A[x][i];
  c[y]=-c[y]*A[x][y];
}
pair<double,vector<double> > simplex( // maximize c^T x s.t. Ax<=b, x>=0
    vector<vector<double> > _A, vector<double> _b, vector<double> _c){
  // returns pair (maximum value, solution vector)
  A=_A;b=_b;c=_c;
  n=b.size();m=c.size();z=0.;
  X=vector<int>(m);Y=vector<int>(n);
  for (int i = 0; i < m; ++i)X[i]=i;
  for (int i = 0; i < n; ++i)Y[i]=i+m;
  while(1){
    int x=-1,y=-1;
    double mn=-EPS;
    for (int i = 0; i < n; ++i)if(b[i]<mn)mn=b[i],x=i;
    if(x<0)break;
    for (int i = 0; i < m; ++i)if(A[x][i]<-EPS){y=i;break;}
    assert(y>=0); // no solution to Ax<=b
    pivot(x,y);
  }
  while(1){
    double mx=EPS;
    int x=-1,y=-1;
    for (int i = 0; i < m; ++i)if(c[i]>mx)mx=c[i],y=i;
    if(y<0)break;
    double mn=1e200;
    for (int i = 0; i < n; ++i)if(A[i][y]>EPS&&b[i]/A[i][y]<mn)mn=b[i]/A
        [i][y],x=i;
    assert(x>=0); // c^T x is unbounded
    pivot(x,y);
  }
  vector<double> r(m);
  for (int i = 0; i < n; ++i)if(Y[i]<m)r[Y[i]]=b[i];
  return {z,r};
}
```

4.21 Discrete log

```
//returns x such that a^x = b (mod m) or -1 if inexistent
ll discrete_log(ll a,ll b,ll m) {
    a%=m, b%=m;
    if(b == 1) return 0;
    int cnt=0;
    ll tmp=1;
    for(int g=__gcd(a,m);g!=1;g=__gcd(a,m)) {
        if(b%g) return -1;
        m/=g, b/=g;
        tmp = tmp*a/g%m;
        ++cnt;
        if(b == tmp) return cnt;
    }
    map<ll,int> w;
    int s = ceil(sqrt(m));
    ll base = b;
    for (int i = 0; i < s; ++i) {
        w[base] = i;
        base=base*a%m;
    }
    base=fastpow(a,s,m);
    ll key=tmp;
    for (int i = 1; i < s+2; ++i) {
        key=base*key%m;
        if(w.count(key)) return i*s-w[key]+cnt;
    }
    return -1;
}
```

4.22 Berlekamp Massey

```
typedef vector<int> vi;
vi BM(vi x){
  vi ls,cur;int lf,ld;
```

```
  for (int i = 0; i < SZ(x); ++i){
    ll t=0;
    for (int j = 0; j < SZ(cur); ++j)t=(t+x[i-j-1]*(ll)cur[j])%MOD;
    if((t-x[i])%MOD==0)continue;
    if(!SZ(cur)){cur.resize(i+1);lf=i;ld=(t-x[i])%MOD;continue;}
    ll k=-(x[i]-t)*fast_pow(ld,MOD-2)%MOD;
    vi c(i-lf-1);c.push_back(k);
    for (int j = 0; j < SZ(ls); ++j)c.push_back(-ls[j]*k%MOD);
    if(SZ(c)<SZ(cur))c.resize(SZ(cur));
    for (int j = 0; j < SZ(cur); ++j)c[j]=(c[j]+cur[j])%MOD;
    if(i-lf+SZ(ls)>=SZ(cur))ls=cur,lf=i,ld=(t-x[i])%MOD;
    cur=c;
  }
  for (int i = 0; i < SZ(cur); ++i)cur[i]=(cur[i]%MOD+MOD)%MOD;
  return cur;
}
```

4.23 Linear Rec

```
//init O(n^2log) query(n^2 logk)
//input: terms: first n term; trans: transition function; MOD; LOG=mxlog
    of k
//output calc(k): kth term mod MOD
//example: {1,1} {2,1} an=2*a_(n-1)+a_(n-2); calc(3)=3 calc(10007)
   =71480733
struct LinearRec{
  typedef vector<int> vi;
  int n; vi terms, trans; vector<vi> bin;
  vi add(vi &a, vi &b){
    vi res(n*2+1);
    for (int i = 0; i < n+1; ++i)for (int j = 0; j < n+1; ++j)res[i+j]=(
        res[i+j]*1LL+(ll)a[i]*b[j])%MOD;
    for(int i=2*n; i>n; --i){
      for (int j = 0; j < n; ++j)res[i-1-j]=(res[i-1-j]*1LL+(ll)res[i]*
          trans[j])%MOD;
      res[i]=0;
    }
    res.erase(res.begin()+n+1,res.end());
    return res;
  }
  LinearRec(vi &terms, vi &trans):terms(terms),trans(trans){
    n=SZ(trans);vi a(n+1);a[1]=1;
    bin.push_back(a);
    for (int i = 1; i < LOG; ++i)bin.push_back(add(bin[i-1],bin[i-1]));
  }
  int calc(int k){
    vi a(n+1);a[0]=1;
    for (int i = 0; i < LOG; ++i)if((k>>i)&1)a=add(a,bin[i]);
    int ret=0;
    for (int i = 0; i < n; ++i)ret=((ll)ret+(ll)a[i+1]*terms[i])%MOD;
    return ret;
  }
};
```

4.24 Tonelli Shanks

```
ll legendre(ll a, ll p){
  if(a%p==0)return 0; if(p==2)return 1;
  return fpow(a,(p-1)/2,p);
}
ll tonelli_shanks(ll n, ll p){  // sqrt(n) mod p (p must be a prime)
  assert(legendre(n,p)==1); if(p==2)return 1;
  ll s=__builtin_ctzll(p-1), q=(p-1LL)>>s, z=rnd(1,p-1);
  if(s==1)return fpow(n,(p+1)/4LL,p);
  while(legendre(z,p)!=p-1)z=rnd(1,p-1);
  ll c=fpow(z,q,p), r=fpow(n,(q+1)/2,p), t=fpow(n,q,p), m=s;
  while(t!=1){
    ll i=1, ts=(t*t)%p;
    while(ts!=1)i++,ts=(ts*ts)%p;
    ll b=c;
    for (int _ = 0; _ < m-i-1; ++_)b=(b*b)%p;
    r=r*b%p;c=b*b%p;t=t*c%p;m=i;
  }
  return r;
}
```

4.25 Prime count

```
// Count number of primes in [1,n]
// O(n^(2/3)) time
// O(sqrt(n)) memory
ll prime_count(ll n) {
    if (n <= 1) return 0;
    int v = sqrtl(n), s = (v+1)>>1, pc=0;
    vector<int> smalls(s), roughs(s), skip(v+1);
    vector<ll> larges(s);
    for (int i = 0; i < s; ++i){
```

```
        smalls[i]=i;
        roughs[i]=2*i+1;
        larges[i]=(n/roughs[i]-1)>>1;
    }
    for(int p=3;p<=v;p+=2) if(!skip[p]){
        int q=p*p;
        if(1ll*q*q>n) break;
        skip[p]=1;
        for(int i=q;i<=v;i+=2*p) skip[i]=1;
        int ns=0;
        for (int k = 0; k < s; ++k){
            int i=roughs[k];
            if(skip[i]) continue;
            ll d=1ll*i*p;
            larges[ns]=larges[k]-(d<=v?larges[smalls[d>>1]-pc]:smalls[(n
                /d-1)>>1])+pc;
            roughs[ns++]=i;
        }
        s=ns;
        for(int i=(v-1)>>1, j=((v/p)-1)|1; j>=p; j-=2){
            int c=smalls[j>>1]-pc;
            for(int e=(j*p)>>1;i>=e;i--) smalls[i]-=c;
        }
        pc++;
    }
    larges[0]+=1ll*(s+2*(pc-1))*(s-1)>>1;
    for (int k = 1; k < s; ++k) larges[0]-=larges[k];
    for (int l = 1; l < s; ++l){
        int q=roughs[l];
        ll m=n/q, t=0;
        int e=smalls[(m/q-1)>>1]-pc;
        if(e<l+1)break;
        for (int k = l+1; k < e+1; ++k) t+=smalls[(m/roughs[k]-1)>>1];
        larges[0]+=t-1ll*(e-l)*(pc+l-1);
    }
    return larges[0]+1;
}
```

5 Geometry

5.1 Point

```
struct pt {  // for 3D add z coordinate
  double x,y;
  pt(double x, double y):x(x),y(y){}
  pt(){}
  double norm2(){return *this**this;}
  double norm(){return sqrt(norm2());}
  bool operator==(pt p){return abs(x-p.x)<=EPS&&abs(y-p.y)<=EPS;}
  pt operator+(pt p){return pt(x+p.x,y+p.y);}
  pt operator-(pt p){return pt(x-p.x,y-p.y);}
  pt operator*(double t){return pt(x*t,y*t);}
  pt operator/(double t){return pt(x/t,y/t);}
  double operator*(pt p){return x*p.x+y*p.y;}
//  pt operator^(pt p){ // only for 3D
//    return pt(y*p.z-z*p.y,z*p.x-x*p.z,x*p.y-y*p.x);}
  double angle(pt p){ // redefine acos for values out of range
    return acos(*this*p/(norm()*p.norm()));}
  pt unit(){return *this/norm();}
  double operator%(pt p){return x*p.y-y*p.x;}
  // 2D from now on
  bool operator<(pt p)const{ // for convex hull
    return x<p.x-EPS||(abs(x-p.x)<=EPS&&y<p.y-EPS);}
  bool left(pt p, pt q){ // is it to the left of directed line pq?
    return (q-p)%(*this-p)>EPS;}
  pt rot(pt r){return pt(*this%r,*this*r);}
  pt rot(double a){return rot(pt(sin(a),cos(a)));}
};
pt ccw90(1,0);
pt cw90(-1,0);
```

5.2 Line

```
int sgn2(double x){return x<0?-1:1;}
struct ln {
  pt p,pq;
  ln(pt p, pt q):p(p),pq(q-p){}
  ln(){}
  bool has(pt r){return dist(r)<=EPS;}
  bool seghas(pt r){return has(r)&&(r-p)*(r-(p+pq))<=EPS;}
//  bool operator /(ln l){return (pq.unit()^l.pq.unit()).norm()<=EPS;}
    // 3D
  bool operator/(ln l){return abs(pq.unit()%l.pq.unit())<=EPS;} // 2D
  bool operator==(ln l){return *this/l&&has(l.p);}
  pt operator^(ln l){ // intersection
    if(*this/l)return pt(DINF,DINF);
```

```
    pt r=l.p+l.pq*((p-l.p)%pq/(l.pq%pq));
//    if(!has(r)){return pt(NAN,NAN,NAN);} // check only for 3D
    return r;
  }
  double angle(ln l){return pq.angle(l.pq);}
  int side(pt r){return has(r)?0:sgn2(pq%(r-p));} // 2D
  pt proj(pt r){return p+pq*((r-p)*pq/pq.norm2());}
  pt ref(pt r){return proj(r)*2-r;}
  double dist(pt r){return (r-proj(r)).norm();}
//  double dist(ln l){ // only 3D
//    if(*this/l)return dist(l.p);
//    return abs((l.p-p)*(pq^l.pq))/(pq^l.pq).norm();
//  }
  ln rot(auto a){return ln(p,p+pq.rot(a));} // 2D
};
ln bisector(ln l, ln m){ // angle bisector
  pt p=l^m;
  return ln(p,p+l.pq.unit()+m.pq.unit());
}
ln bisector(pt p, pt q){ // segment bisector (2D)
  return ln((p+q)*.5,p).rot(ccw90);
}
```

5.3 Circle

```
struct circle {
  pt o;double r;
  circle(pt o, double r):o(o),r(r){}
  circle(pt x, pt y, pt z){o=bisector(x,y)^bisector(x,z);r=(o-x).norm()
      ;}
  bool has(pt p){return (o-p).norm()<=r+EPS;}
  vector<pt> operator^(circle c){ // ccw
    vector<pt> s;
    double d=(o-c.o).norm();
    if(d>r+c.r+EPS||d+min(r,c.r)+EPS<max(r,c.r))return s;
    double x=(d*d-c.r*c.r+r*r)/(2*d);
    double y=sqrt(r*r-x*x);
    pt v=(c.o-o)/d;
    s.push_back(o+v*x-v.rot(ccw90)*y);
    if(y>EPS)s.push_back(o+v*x+v.rot(ccw90)*y);
    return s;
  }
  vector<pt> operator^(ln l){
    vector<pt> s;
    pt p=l.proj(o);
    double d=(p-o).norm();
    if(d-EPS>r)return s;
    if(abs(d-r)<=EPS){s.push_back(p);return s;}
    d=sqrt(r*r-d*d);
    s.push_back(p+l.pq.unit()*d);
    s.push_back(p-l.pq.unit()*d);
    return s;
  }
  vector<pt> tang(pt p){
    double d=sqrt((p-o).norm2()-r*r);
    return *this^circle(p,d);
  }
  bool in(circle c){ // non strict
    double d=(o-c.o).norm();
    return d+r<=c.r+EPS;
  }
  double intertriangle(pt a, pt b){ // area of intersection with oab
    if(abs((o-a)%(o-b))<=EPS)return 0.;
    vector<pt> q={a},w=*this^ln(a,b);
    if(w.size()==2)for(auto p:w)if((a-p)*(b-p)<-EPS)q.push_back(p);
    q.push_back(b);
    if(q.size()==4&&(q[0]-q[1])*(q[2]-q[1])>EPS)swap(q[1],q[2]);
    double s=0;
    for (int i = 0; i < q.size()-1; ++i){
      if(!has(q[i])||!has(q[i+1]))s+=r*r*(q[i]-o).angle(q[i+1]-o)/2;
      else s+=abs((q[i]-o)%(q[i+1]-o)/2);
    }
    return s;
  }
};
vector<double> intercircles(vector<circle> c){
  vector<double> r(SZ(c)+1); // r[k]: area covered by at least k circles
  for (int i = 0; i < SZ(c); ++i){          // O(n^2 log n) (high
      constant)
    int k=1;Cmp s(c[i].o);
    vector<pair<pt,int> > p={
      {c[i].o+pt(1,0)*c[i].r,0},
      {c[i].o-pt(1,0)*c[i].r,0}};
    for (int j = 0; j < SZ(c); ++j)if(j!=i){
      bool b0=c[i].in(c[j]),b1=c[j].in(c[i]);
      if(b0&&(!b1||i<j))k++;
```

```
      else if(!b0&&!b1){
        auto v=c[i]^c[j];
        if(SZ(v)==2){
          p.push_back({v[0],1});p.push_back({v[1],-1});
          if(s(v[1],v[0]))k++;
        }
      }
    }
    sort(p.begin(),p.end(),
      [&](pair<pt,int> a, pair<pt,int> b){return s(a.first,b.first);});
    for (int j = 0; j < SZ(p); ++j){
      pt p0=p[j?j-1:SZ(p)-1].first,p1=p[j].first;
      double a=(p0-c[i].o).angle(p1-c[i].o);
      r[k]+=(p0.x-p1.x)*(p0.y+p1.y)/2+c[i].r*c[i].r*(a-sin(a))/2;
      k+=p[j].second;
    }
  }
  return r;
}
```

5.4 Polygon

```
int sgn(double x){return x<-EPS?-1:x>EPS;}
struct pol {
  int n;vector<pt> p;
  pol(){}
  pol(vector<pt> _p){p=_p;n=p.size();}
  double area(){
    double r=0.;
    for (int i = 0; i < n; ++i)r+=p[i]%p[(i+1)%n];
    return abs(r)/2; // negative if CW, positive if CCW
  }
  pt centroid(){ // (barycenter)
    pt r(0,0);double t=0;
    for (int i = 0; i < n; ++i){
      r=r+(p[i]+p[(i+1)%n])*(p[i]%p[(i+1)%n]);
      t+=p[i]%p[(i+1)%n];
    }
    return r/t/3;
  }
  bool has(pt q){ // O(n)
    for (int i = 0; i < n; ++i)if(ln(p[i],p[(i+1)%n]).seghas(q))return
        true;
    int cnt=0;
    for (int i = 0; i < n; ++i){
      int j=(i+1)%n;
      int k=sgn((q-p[j])%(p[i]-p[j]));
      int u=sgn(p[i].y-q.y),v=sgn(p[j].y-q.y);
      if(k>0&&u<0&&v>=0)cnt++;
      if(k<0&&v<0&&u>=0)cnt--;
    }
    return cnt!=0;
  }
  void normalize(){ // (call before haslog, remove collinear first)
    if(p[2].left(p[0],p[1]))reverse(p.begin(),p.end());
    int pi=min_element(p.begin(),p.end())-p.begin();
    vector<pt> s(n);
    for (int i = 0; i < n; ++i)s[i]=p[(pi+i)%n];
    p.swap(s);
  }
  bool haslog(pt q){ // O(log(n)) only CONVEX. Call normalize first
    if(q.left(p[0],p[1])||q.left(p.back(),p[0]))return false;
    int a=1,b=p.size()-1;  // returns true if point on boundary
    while(b-a>1){          // (change sign of EPS in left
      int c=(a+b)/2;       //  to return false in such case)
      if(!q.left(p[0],p[c]))a=c;
      else b=c;
    }
    return !q.left(p[a],p[a+1]);
  }
  pt farthest(pt v){ // O(log(n)) only CONVEX
    if(n<10){
      int k=0;
      for (int i = 1; i < n; ++i)if(v*(p[i]-p[k])>EPS)k=i;
      return p[k];
    }
    if(n==SZ(p))p.push_back(p[0]);
    pt a=p[1]-p[0];
    int s=0,e=n,ua=v*a>EPS;
    if(!ua&&v*(p[n-1]-p[0])<=EPS)return p[0];
    while(1){
      int m=(s+e)/2;pt c=p[m+1]-p[m];
      int uc=v*c>EPS;
      if(!uc&&v*(p[m-1]-p[m])<=EPS)return p[m];
      if(ua&&(!uc||v*(p[s]-p[m])>EPS))e=m;
      else if(ua||uc||v*(p[s]-p[m])>=-EPS)s=m,a=c,ua=uc;
```

```
      else e=m;
      assert(e>s+1);
    }
  }
  pol cut(ln l){   // cut CONVEX polygon by line l
    vector<pt> q;  // returns part at left of l.pq
    for (int i = 0; i < n; ++i){
      int d0=sgn(l.pq%(p[i]-l.p)),d1=sgn(l.pq%(p[(i+1)%n]-l.p));
      if(d0>=0)q.push_back(p[i]);
      ln m(p[i],p[(i+1)%n]);
      if(d0*d1<0&&!(l/m))q.push_back(l^m);
    }
    return pol(q);
  }
  double intercircle(circle c){ // area of intersection with circle
    double r=0.;
    for (int i = 0; i < n; ++i){
      int j=(i+1)%n;double w=c.intertriangle(p[i],p[j]);
      if((p[j]-c.o)%(p[i]-c.o)>0)r+=w;
      else r-=w;
    }
    return abs(r);
  }
  double callipers(){ // square distance of most distant points
    double r=0;     // prereq: convex, ccw, NO COLLINEAR POINTS
    for(int i=0,j=n<2?0:1;i<j;++i){
      for(;;j=(j+1)%n){
        r=max(r,(p[i]-p[j]).norm2());
        if((p[(i+1)%n]-p[i])%(p[(j+1)%n]-p[j])<=EPS)break;
      }
    }
    return r;
  }
};
// Dynamic convex hull trick
vector<pol> w;
void add(pt q){ // add(q), O(log^2(n))
  vector<pt> p={q};
  while(!w.empty()&&SZ(w.back().p)<2*SZ(p)){
    for(pt v:w.back().p)p.push_back(v);
    w.pop_back();
  }
  w.push_back(pol(chull(p)));
}
ll query(pt v){ // max(q*v:q in w), O(log^2(n))
  ll r=-INF;
  for(auto& p:w)r=max(r,p.farthest(v)*v);
  return r;
}
```

5.5 Plane

```
struct plane {
  pt a,n; // n: normal unit vector
  plane(pt a, pt b, pt c):a(a),n(((b-a)^(c-a)).unit()){}
  plane(){}
  bool has(pt p){return abs((p-a)*n)<=EPS;}
  double angle(plane w){return acos(n*w.n);}
  double dist(pt p){return abs((p-a)*n);}
  pt proj(pt p){inter(ln(p,p+n),p);return p;}
  bool inter(ln l, pt& r){
    double x=n*(l.p+l.pq-a),y=n*(l.p-a);
    if(abs(x-y)<=EPS)return false;
    r=(l.p*x-(l.p+l.pq)*y)/(x-y);
    return true;
  }
  bool inter(plane w, ln& r){
    pt nn=n^w.n;pt v=n^nn;double d=w.n*v;
    if(abs(d)<=EPS)return false;
    pt p=a+v*(w.n*(w.a-a)/d);
    r=ln(p,p+nn);
    return true;
  }
};
```

5.6 Radial order of points

```
struct Cmp { // IMPORTANT: add const in pt operator -
  pt r;
  Cmp(pt r):r(r){}
  int cuad(const pt &a)const {
    if(a.x>0&&a.y>=0)return 0;
    if(a.x<=0&&a.y>0)return 1;
    if(a.x<0&&a.y<=0)return 2;
    if(a.x>=0&&a.y<0)return 3;
    assert(a.x==0&&a.y==0);
    return -1;
```

```
  }
  bool cmp(const pt& p1, const pt& p2)const {
    int c1=cuad(p1),c2=cuad(p2);
    if(c1==c2)return p1.y*p2.x<p1.x*p2.y;
    return c1<c2;
  }
  bool operator()(const pt& p1, const pt& p2)const {
    return cmp(p1-r,p2-r);
  }
};
```

5.7 Convex hull

```
// CCW order
// Includes collinear points (change sign of EPS in left to exclude)
vector<pt> chull(vector<pt> p){
  vector<pt> r;
  sort(p.begin(),p.end()); // first x, then y
  p.erase(unique(ALL(p)), p.end());
  for (int i = 0; i < p.size(); ++i){ // lower hull
    while(r.size()>=2&&r.back().left(r[r.size()-2],p[i]))r.pop_back();
    r.push_back(p[i]);
  }
  r.pop_back();
  int k=r.size(),pr=k-1;
  for(int i=p.size()-1;i>=0;--i){ // upper hull
    while(r.size()>=k+2&&r.back().left(r[r.size()-2],p[i]))r.pop_back();
    while(pr>=0&&p[i]<r[pr]) pr--;
    if(pr<0||!(p[i]==r[pr])) r.push_back(p[i]);
  }
  return r;
}
```

5.8 Dual from planar graph

```
vector<int> g[MAXN];int n; // input graph (must be connected)
vector<int> gd[MAXN];int nd; // output graph
vector<int> nodes[MAXN]; // nodes delimiting region (in CW order)
map<pair<int,int>,int> ps,es;
void get_dual(vector<pt> p){ // p: points corresponding to nodes
  ps.clear();es.clear();
  for (int x = 0; x < n; ++x){
    Cmp pc(p[x]); // (radial order of points)
    auto comp=[&](int a, int b){return pc(p[a],p[b]);};
    sort(g[x].begin(),g[x].end(),comp);
    for (int i = 0; i < g[x].size(); ++i)ps[{x,g[x][i]}]=i;
  }
  nd=0;
  for (int xx = 0; xx < n; ++xx)for(auto yy:g[xx])if(!es.count({xx,yy}))
      {
    int x=xx,y=yy;gd[nd].clear();nodes[nd].clear();
    while(!es.count({x,y})){
      es[{x,y}]=nd;nodes[nd].push_back(y);
      int z=g[y][(ps[{y,x}]+1)%g[y].size()];x=y;y=z;
    }
    nd++;
  }
  for(auto p:es){
    pair<int,int> q={p.first.second,p.first.first};
    assert(es.count(q));
    if(es[q]!=p.second)gd[p.second].push_back(es[q]);
  }
  for (int i = 0; i < nd; ++i){
    sort(gd[i].begin(),gd[i].end());
    gd[i].erase(unique(gd[i].begin(),gd[i].end()),gd[i].end());
  }
}
```

5.9 Halfplane intersection

```
// polygon intersecting left side of halfplanes
struct halfplane:public ln{
  double angle;
  halfplane(){}
  halfplane(pt a,pt b){p=a; pq=b-a; angle=atan2(pq.y,pq.x);}
  bool operator<(halfplane b)const{return angle<b.angle;}
  bool out(pt q){return pq%(q-p)<-EPS;}
};
vector<pt> intersect(vector<halfplane> b){
  vector<pt>bx={{DINF,DINF},{-DINF,DINF},{-DINF,-DINF},{DINF,-DINF}};
  for (int i = 0; i < 4; ++i) b.push_back(halfplane(bx[i],bx[(i+1)%4]));
  sort(ALL(b));
  int n=SZ(b),q=1,h=0;
  vector<halfplane> c(SZ(b)+10);
  for (int i = 0; i < n; ++i){
    while(q<h&&b[i].out(c[h]^c[h-1])) h--;
    while(q<h&&b[i].out(c[q]^c[q+1])) q++;
```

```
    c[++h]=b[i];
    if(q<h&&abs(c[h].pq%c[h-1].pq)<EPS){
      if(c[h].pq*c[h-1].pq<=0) return {};
      h--;
      if(b[i].out(c[h].p)) c[h]=b[i];
    }
  }
  while(q<h-1&&c[q].out(c[h]^c[h-1]))h--;
  while(q<h-1&&c[h].out(c[q]^c[q+1]))q++;
  if(h-q<=1)return {};
  c[h+1]=c[q];
  vector<pt> s;
  for (int i = q; i < h+1; ++i) s.push_back(c[i]^c[i+1]);
  return s;
}
```

5.10 KD tree

```
// given a set of points, answer queries of nearest point in O(log(n))
bool onx(pt a, pt b){return a.x<b.x;}
bool ony(pt a, pt b){return a.y<b.y;}
struct Node {
  pt pp;
  ll x0=INF, x1=-INF, y0=INF, y1=-INF;
  Node *first=0, *second=0;
  ll distance(pt p){
    ll x=min(max(x0,p.x),x1);
    ll y=min(max(y0,p.y),y1);
    return (pt(x,y)-p).norm2();
  }
  Node(vector<pt>&& vp):pp(vp[0]){
    for(pt p:vp){
      x0=min(x0,p.x); x1=max(x1,p.x);
      y0=min(y0,p.y); y1=max(y1,p.y);
    }
    if(SZ(vp)>1){
      sort(ALL(vp),x1-x0>=y1-y0?onx:ony);
      int m=SZ(vp)/2;
      first=new Node({vp.begin(),vp.begin()+m});
      second=new Node({vp.begin()+m,vp.end()});
    }
  }
};
```

```
struct KDTree {
  Node* root;
  KDTree(const vector<pt>& vp):root(new Node({ALL(vp)})) {}
  pair<ll,pt> search(pt p, Node *node){
    if(!node->first){
      //avoid query point as answer
      //if(p==node->pp) {INF,pt()};
      return {(p-node->pp).norm2(),node->pp};
    }
    Node *f=node->first, *s=node->second;
    ll bf=f->distance(p), bs=s->distance(p);
    if(bf>bs)swap(bf,bs),swap(f,s);
    auto best=search(p,f);
    if(bs<best.first) best=min(best,search(p,s));
    return best;
  }
  pair<ll,pt> nearest(pt p){return search(p,root);}
};
```

5.11 Minkowski sum

```
// Compute Minkowski sum of two CONVEX polygons (remove collinear first)
// Returns answer in CCW order
void reorder(vector<pt> &p){
  if(!p[2].left(p[0],p[1])) reverse(ALL(p));
  int pos=0;
  for (int i = 1; i < SZ(p); ++i) if(pt(p[i].y,p[i].x) < pt(p[pos].y,p[
      pos].x)) pos=i;
  rotate(p.begin(), p.begin()+pos, p.end());
}
vector<pt> minkowski_sum(vector<pt> p, vector<pt> q){
  if(min(SZ(p),SZ(q))<3){
    vector<pt> v;
    for(pt pp:p) for(pt qq:q) v.push_back(pp+qq);
    return chull(v);
  }
  reorder(p); reorder(q);
  for (int i = 0; i < 2; ++i) p.push_back(p[i]), q.push_back(q[i]);
  vector<pt> r;
  int i=0,j=0;
  while(i+2<SZ(p)||j+2<SZ(q)){
    r.push_back(p[i]+q[j]);
    auto cross=(p[i+1]-p[i])%(q[j+1]-q[j]);
```

```
    i+=cross>=-EPS;
    j+=cross<=EPS;
  }
  return r;
}
```

5.12 Delaunay triangulation

```
// Returns planar graph representing Delaunay's triangulation.
// Edges for each vertex are in ccw order.
// It can work with doubles, but also integers (replace long double in line 51)
typedef struct QuadEdge* Q;
struct QuadEdge {
  int id,used;
  pt o;
  Q rot,nxt;
  QuadEdge(int id=-1, pt o=pt(INF,INF)):id(id),used(0),o(o),rot(0),nxt(0){}
  Q rev(){return rot->rot;}
  Q next(){return nxt;}
  Q prev(){return rot->next()->rot;}
  pt dest(){return rev()->o;}
};

Q edge(pt a, pt b, int ida, int idb){
  Q e1=new QuadEdge(ida,a);
  Q e2=new QuadEdge(idb,b);
  Q e3=new QuadEdge;
  Q e4=new QuadEdge;
  tie(e1->rot,e2->rot,e3->rot,e4->rot)={e3,e4,e2,e1};
  tie(e1->nxt,e2->nxt,e3->nxt,e4->nxt)={e1,e2,e4,e3};
  return e1;
}

void splice(Q a, Q b){
  swap(a->nxt->rot->nxt,b->nxt->rot->nxt);
  swap(a->nxt,b->nxt);
}

void del_edge(Q& e, Q ne){
  splice(e,e->prev()); splice(e->rev(),e->rev()->prev());
  delete e->rev()->rot; delete e->rev();
  delete e->rot; delete e;
  e=ne;
}

Q conn(Q a, Q b){
  Q e=edge(a->dest(),b->o,a->rev()->id,b->id);
  splice(e,a->rev()->prev());
  splice(e->rev(),b);
  return e;
}

auto area(pt p, pt q, pt r){return (q-p)%(r-q);}

// is p in circunference formed by (a,b,c)?
bool in_c(pt a, pt b, pt c, pt p){
  // Warning: this number is O(max_coord^4).
  // Consider using int128 or using an alternative method for this function
  long double p2=p*p,A=a*a-p2,B=b*b-p2,C=c*c-p2;
  return area(p,a,b)*C+area(p,b,c)*A+area(p,c,a)*B>EPS;
}

pair<Q,Q> build_tr(vector<pt>& p, int l, int r){
  if(r-l+1<=3){
    Q a=edge(p[l],p[l+1],l,l+1),b=edge(p[l+1],p[r],l+1,r);
    if(r-l+1==2) return {a,a->rev()};
    splice(a->rev(),b);
    auto ar=area(p[l],p[l+1],p[r]);
    Q c=abs(ar)>EPS?conn(b,a):0;
    if(ar>=-EPS) return {a,b->rev()};
    return {c->rev(),c};
  }
  int m=(l+r)/2;
  auto [la,ra]=build_tr(p,l,m);
  auto [lb,rb]=build_tr(p,m+1,r);
  while(1){
    if(ra->dest().left(lb->o,ra->o)) ra=ra->rev()->prev();
    else if(lb->dest().left(lb->o,ra->o)) lb=lb->rev()->next();
    else break;
  }
  Q b=conn(lb->rev(),ra);
  auto valid=[&](Q e){return b->o.left(e->dest(),b->dest());};
  if(ra->o==la->o) la=b->rev();
```

```
  if(lb->o==rb->o) rb=b;
  while(1){
    Q L=b->rev()->next();
    if(valid(L)) while(in_c(b->dest(),b->o,L->dest(),L->next()->dest()))
        del_edge(L,L->next());
    Q R=b->prev();
    if(valid(R)) while(in_c(b->dest(),b->o,R->dest(),R->prev()->dest()))
        del_edge(R,R->prev());
    if(!valid(L)&&!valid(R)) break;
    if(!valid(L)||(valid(R)&&in_c(L->dest(),L->o,R->o,R->dest()))) b=
        conn(R,b->rev());
    else b=conn(b->rev(),L->rev());
  }
  return {la,rb};
}

vector<vector<int>> delaunay(vector<pt> v){
  int n=SZ(v); auto tmp=v;
  vector<int> id(n); iota(ALL(id),0);
  sort(ALL(id),[&](int l, int r){return v[l]<v[r];});
  for (int i = 0; i < n; ++i) v[i]=tmp[id[i]];
  assert(unique(ALL(v))==v.end());
  vector<vector<int>> g(n);
  int col=1;
  for (int i = 2; i < n; ++i) col&=abs(area(v[i],v[i-1],v[i-2]))<=EPS;
  if(col){
    for (int i = 1; i < n; ++i) g[id[i-1]].push_back(id[i]),g[id[i]].
        push_back(id[i-1]);
  }
  else{
    Q e=build_tr(v,0,n-1).first;
    vector<Q> edg={e};
    for(int i=0;i<SZ(edg);e=edg[i++]){
      for(Q at=e;!at->used;at=at->next()){
        at->used=1;
        g[id[at->id]].push_back(id[at->rev()->id]);
        edg.push_back(at->rev());
      }
    }
  }
  return g;
}
```

6 Strings

6.1 KMP

```
vector<int> kmppre(string& t){ // r[i]: longest border of t[0,i)
  vector<int> r(t.size()+1);r[0]=-1;
  int j=-1;
  for (int i = 0; i < t.size(); ++i){
    while(j>=0&&t[i]!=t[j])j=r[j];
    r[i+1]=++j;
  }
  return r;
}
void kmp(string& s, string& t){ // find t in s
  int j=0;vector<int> b=kmppre(t);
  for (int i = 0; i < s.size(); ++i){
    while(j>=0&&s[i]!=t[j])j=b[j];
    if(++j==t.size())printf("Match at %d\n",i-j+1),j=b[j];
  }
}
```

6.2 Z function

```
vector<int> z_function(string& s){
  int l=0,r=0,n=s.size();
  vector<int> z(s.size(),0); // z[i] = max k: s[0,k) == s[i,i+k)
  for (int i = 1; i < n; ++i){
    if(i<=r)z[i]=min(r-i+1,z[i-l]);
    while(i+z[i]<n&&s[z[i]]==s[i+z[i]])z[i]++;
    if(i+z[i]-1>r)l=i,r=i+z[i]-1;
  }
  return z;
}
```

6.3 Manacher

```
int d1[MAXN];//d1[i] = max odd palindrome centered on i
int d2[MAXN];//d2[i] = max even palindrome centered on i
//s  aabbaacaabbaa
//d1 1111117111111
//d2 0103010010301
void manacher(string& s){
  int l=0,r=-1,n=s.size();
  for (int i = 0; i < n; ++i){
```

```
    int k=i>r?1:min(d1[l+r-i],r-i);
    while(i+k<n&&i-k>=0&&s[i+k]==s[i-k])k++;
    d1[i]=k--;
    if(i+k>r)l=i-k,r=i+k;
  }
  l=0;r=-1;
  for (int i = 0; i < n; ++i){
    int k=i>r?0:min(d2[l+r-i+1],r-i+1);k++;
    while(i+k<=n&&i-k>=0&&s[i+k-1]==s[i-k])k++;
    d2[i]=--k;
    if(i+k-1>r)l=i-k,r=i+k-1;
  }
}
```

6.4 Aho-Corasick

```
struct vertex {
  map<char,int> next,go;
  int p,link;
  char pch;
  vector<int> leaf;
  vertex(int p=-1, char pch=-1):p(p),pch(pch),link(-1){}
};
vector<vertex> t;
void aho_init(){ //do not forget!!
  t.clear();t.push_back(vertex());
}
void add_string(string s, int id){
  int v=0;
  for(char c:s){
    if(!t[v].next.count(c)){
      t[v].next[c]=t.size();
      t.push_back(vertex(v,c));
    }
    v=t[v].next[c];
  }
  t[v].leaf.push_back(id);
}
int go(int v, char c);
int get_link(int v){
  if(t[v].link<0)
    if(!v||!t[v].p)t[v].link=0;
    else t[v].link=go(get_link(t[v].p),t[v].pch);
  return t[v].link;
}
int go(int v, char c){
  if(!t[v].go.count(c))
    if(t[v].next.count(c))t[v].go[c]=t[v].next[c];
    else t[v].go[c]=v==0?0:go(get_link(v),c);
  return t[v].go[c];
}
```

6.5 Suffix automaton

```
struct state {int len,link;map<char,int> next;}; //clear next!!
state st[100005];
int sz,last;
void sa_init(){
  last=st[0].len=0;sz=1;
  st[0].link=-1;
}
void sa_extend(char c){
  int k=sz++,p;
  st[k].len=st[last].len+1;
  for(p=last;p!=-1&&!st[p].next.count(c);p=st[p].link)st[p].next[c]=k;
  if(p==-1)st[k].link=0;
  else {
    int q=st[p].next[c];
    if(st[p].len+1==st[q].len)st[k].link=q;
    else {
      int w=sz++;
      st[w].len=st[p].len+1;
      st[w].next=st[q].next;st[w].link=st[q].link;
      for(;p!=-1&&st[p].next[c]==q;p=st[p].link)st[p].next[c]=w;
      st[q].link=st[k].link=w;
    }
  }
  last=k;
}
//  input: abcbcbc
//  i,link,len,next
//  0 -1 0 (a,1) (b,5) (c,7)
//  1 0 1 (b,2)
//  2 5 2 (c,3)
//  3 7 3 (b,4)
//  4 9 4 (c,6)
```

```
//  5 0 1 (c,7)
//  6 11 5 (b,8)
//  7 0 2 (b,9)
//  8 9 6 (c,10)
//  9 5 3 (c,11)
//  10 11 7
//  11 7 4 (b,8)
```

6.6 Palindromic Tree

```
struct palindromic_tree{
    static const int SIGMA=26;
    struct Node{
        int len, link, to[SIGMA];
        ll cnt;
        Node(int len, int link=0, ll cnt=1):len(len),link(link),cnt(cnt)
            {
            memset(to,0,sizeof(to));
        }
    };
    vector<Node> ns;
    int last;
    palindromic_tree():last(0){ns.push_back(Node(-1));ns.push_back(Node
        (0));}
    void add(int i, string &s){
        int p=last, c=s[i]-'a';
        while(s[i-ns[p].len-1]!=s[i])p=ns[p].link;
        if(ns[p].to[c]){
            last=ns[p].to[c];
            ns[last].cnt++;
        }else{
            int q=ns[p].link;
            while(s[i-ns[q].len-1]!=s[i])q=ns[q].link;
            q=max(1,ns[q].to[c]);
            last=ns[p].to[c]=SZ(ns);
            ns.push_back(Node(ns[p].len+2,q,1));
        }
    }
};
```

6.7 Suffix array (shorter but slower)

```
pair<int, int> sf[MAXN];
bool sacomp(int lhs, int rhs) {return sf[lhs]<sf[rhs];}
vector<int> constructSA(string& s){ // O(n log^2(n))
  int n=s.size();                   // (sometimes fast enough)
  vector<int> sa(n),r(n);
  for (int i = 0; i < n; ++i)r[i]=s[i];
  for(int m=1;m<n;m*=2){
    for (int i = 0; i < n; ++i)sa[i]=i,sf[i]={r[i],i+m<n?r[i+m]:-1};
    stable_sort(sa.begin(),sa.end(),sacomp);
    r[sa[0]]=0;
    for (int i = 1; i < n; ++i)r[sa[i]]=sf[sa[i]]!=sf[sa[i-1]]?i:r[sa[i
        -1]];
  }
  return sa;
}
```

6.8 Suffix array

```
#define RB(x) (x<n?r[x]:0)
void csort(vector<int>& sa, vector<int>& r, int k){
  int n=sa.size(),top=max(255,n+1);
  vector<int> f(top,0),t(n);
  for (int i = 0; i < n; ++i)f[RB(i+k)]++;
  int sum=0;
  for (int i = 0; i < top; ++i)f[i]=(sum+=f[i])-f[i];
  for (int i = 0; i < n; ++i)t[f[RB(sa[i]+k)]++]=sa[i];
  sa=t;
}
vector<int> constructSA(string& s){ // O(n logn)
  int n=s.size(),rank;
  vector<int> sa(n),r(n),t(n);
  for (int i = 0; i < n; ++i)sa[i]=i,r[i]=s[i];
  for(int k=1;k<n;k*=2){
    csort(sa,r,k);csort(sa,r,0);
    t[sa[0]]=rank=1;
    for (int i = 1; i < n; ++i){
      if(r[sa[i]]!=r[sa[i-1]]||RB(sa[i]+k)!=RB(sa[i-1]+k))rank++;
      t[sa[i]]=rank;
    }
    r=t;
    if(r[sa[n-1]]==n)break;
  }
  return sa;
}
```

6.9 LCP (Longest Common Prefix)

```
vector<int> computeLCP(string& s, vector<int>& sa){
  int n=s.size(),L=0;
  vector<int> lcp(n),plcp(n),phi(n);
  phi[sa[0]]=-1;
  for (int i = 1; i < n; ++i)phi[sa[i]]=sa[i-1];
  for (int i = 0; i < n; ++i){
    if(phi[i]<0){plcp[i]=0;continue;}
    while(s[i+L]==s[phi[i]+L])L++;
    plcp[i]=L;
    L=max(L-1,0);
  }
  for (int i = 0; i < n; ++i)lcp[i]=plcp[sa[i]];
  return lcp; // lcp[i]=LCP(sa[i-1],sa[i])
}
```

6.10 Suffix Tree (Ukkonen's algorithm)

```
struct SuffixTree {
  char s[MAXN];
  map<int,int> to[MAXN];
  int len[MAXN]={INF},fpos[MAXN],link[MAXN];
  int node,pos,sz=1,n=0;
  int make_node(int p, int l){
    fpos[sz]=p;len[sz]=l;return sz++;}
  void go_edge(){
    while(pos>len[to[node][s[n-pos]]]){
      node=to[node][s[n-pos]];
      pos-=len[node];
    }
  }
  void add(int c){
    s[n++]=c;pos++;
    int last=0;
    while(pos>0){
      go_edge();
      int edge=s[n-pos];
      int& v=to[node][edge];
      int t=s[fpos[v]+pos-1];
      if(v==0){
        v=make_node(n-pos,INF);
        link[last]=node;last=0;
      }
      else if(t==c){link[last]=node;return;}
      else {
        int u=make_node(fpos[v],pos-1);
        to[u][c]=make_node(n-1,INF);
        to[u][t]=v;
        fpos[v]+=pos-1;len[v]-=pos-1;
        v=u;link[last]=u;last=u;
      }
      if(node==0)pos--;
      else node=link[node];
    }
  }
};
```

6.11 Hashing

```
struct Hash {
  int P=1777771,MOD[2],PI[2];
  vector<int> h[2],pi[2];
  Hash(string& s){
    MOD[0]=999727999;MOD[1]=1070777777;
    PI[0]=325255434;PI[1]=10018302;
    for (int k = 0; k < 2; ++k)h[k].resize(s.size()+1),pi[k].resize(s.
        size()+1);
    for (int k = 0; k < 2; ++k){
      h[k][0]=0;pi[k][0]=1;
      ll p=1;
      for (int i = 1; i < s.size()+1; ++i){
        h[k][i]=(h[k][i-1]+p*s[i-1])%MOD[k];
        pi[k][i]=(1LL*pi[k][i-1]*PI[k])%MOD[k];
        p=(p*P)%MOD[k];
      }
    }
  }
  ll get(int s, int e){
    ll h0=(h[0][e]-h[0][s]+MOD[0])%MOD[0];
    h0=(1LL*h0*pi[0][s])%MOD[0];
    ll h1=(h[1][e]-h[1][s]+MOD[1])%MOD[1];
    h1=(1LL*h1*pi[1][s])%MOD[1];
    return (h0<<32)|h1;
  }
};
```

6.12 Hashing with ll (using __int128)

```cpp
#define bint __int128
struct Hash {
  bint MOD=2123456789876543211LL,P=1777771,PI=10695574108965957 1LL;
  vector<bint> h,pi;
  Hash(string& s){
    assert((P*PI)%MOD==1);
    h.resize(s.size()+1);pi.resize(s.size()+1);
    h[0]=0;pi[0]=1;
    bint p=1;
    for (int i = 1; i < s.size()+1; ++i){
      h[i]=(h[i-1]+p*s[i-1])%MOD;
      pi[i]=(pi[i-1]*PI)%MOD;
      p=(p*P)%MOD;
    }
  }
  ll get(int s, int e){
    return (((h[e]-h[s]+MOD)%MOD)*pi[s])%MOD;
  }
};
```

7 Flow

7.1 Matching (slower)

```cpp
vector<int> g[MAXN]; // [0,n)->[0,m)
int n,m;
int mat[MAXM];bool vis[MAXN];
int match(int x){
  if(vis[x])return 0;
  vis[x]=true;
  for(int y:g[x])if(mat[y]<0||match(mat[y])){mat[y]=x;return 1;}
  return 0;
}
vector<pair<int,int> > max_matching(){
  vector<pair<int,int> > r;
  memset(mat,-1,sizeof(mat));
  for (int i = 0; i < n; ++i)memset(vis,false,sizeof(vis)),match(i);
  for (int i = 0; i < m; ++i)if(mat[i]>=0)r.push_back({mat[i],i});
  return r;
}
```

7.2 Matching (Hopcroft-Karp)

```cpp
vector<int> g[MAXN]; // [0,n)->[0,m)
int n,m;
int mt[MAXN],mt2[MAXN],ds[MAXN];
bool bfs(){
  queue<int> q;
  memset(ds,-1,sizeof(ds));
  for (int i = 0; i < n; ++i)if(mt2[i]<0)ds[i]=0,q.push(i);
  bool r=false;
  while(!q.empty()){
    int x=q.front();q.pop();
    for(int y:g[x]){
      if(mt[y]>=0&&ds[mt[y]]<0)ds[mt[y]]=ds[x]+1,q.push(mt[y]);
      else if(mt[y]<0)r=true;
    }
  }
  return r;
}
bool dfs(int x){
  for(int y:g[x])if(mt[y]<0||ds[mt[y]]==ds[x]+1&&dfs(mt[y])){
    mt[y]=x;mt2[x]=y;
    return true;
  }
  ds[x]=1<<30;
  return false;
}
int mm(){
  int r=0;
  memset(mt,-1,sizeof(mt));memset(mt2,-1,sizeof(mt2));
  while(bfs()){
    for (int i = 0; i < n; ++i)if(mt2[i]<0)r+=dfs(i);
  }
  return r;
}
```

7.3 Hungarian

```cpp
typedef long double td; typedef vector<int> vi; typedef vector<td> vd;
const td INF=1e100;//for maximum set INF to 0, and negate costs
bool zero(td x){return fabs(x)<1e-9;}//change to x==0, for ints/ll
struct Hungarian{
    int n; vector<vd> cs; vi L, R;
```

```
    Hungarian(int N, int M):n(max(N,M)),cs(n,vd(n)),L(n),R(n){
        for (int x = 0; x < N; ++x)for (int y = 0; y < M; ++y)cs[x][y]=
            INF;
    }
    void set(int x,int y,td c){cs[x][y]=c;}
  td assign() {
    int mat = 0; vd ds(n), u(n), v(n); vi dad(n), sn(n);
    for (int i = 0; i < n; ++i)u[i]=*min_element(ALL(cs[i]));
    for (int j = 0; j < n; ++j){v[j]=cs[0][j]-u[0];for (int i = 1; i < n
        ; ++i)v[j]=min(v[j],cs[i][j]-u[i]);}
    L=R=vi(n, -1);
    for (int i = 0; i < n; ++i)for (int j = 0; j < n; ++j)
      if(R[j]==-1&&zero(cs[i][j]-u[i]-v[j])){L[i]=j;R[j]=i;mat++;break;}
    for(;mat<n;mat++){
        int s=0, j=0, i;
        while(L[s] != -1)s++;
        fill(ALL(dad),-1);fill(ALL(sn),0);
        for (int k = 0; k < n; ++k)ds[k]=cs[s][k]-u[s]-v[k];
        for(;;){
            j = -1;
            for (int k = 0; k < n; ++k)if(!sn[k]&&(j==-1||ds[k]<ds[j]))j
                =k;
            sn[j] = 1; i = R[j];
            if(i == -1) break;
            for (int k = 0; k < n; ++k)if(!sn[k]){
                auto new_ds=ds[j]+cs[i][k]-u[i]-v[k];
                if(ds[k] > new_ds){ds[k]=new_ds;dad[k]=j;}
            }
        }
        for (int k = 0; k < n; ++k)if(k!=j&&sn[k]){auto w=ds[k]-ds[j];v[
            k]+=w,u[R[k]]-=w;}
        u[s] += ds[j];
        while(dad[j]>=0){int d = dad[j];R[j]=R[d];L[R[j]]=j;j=d;}
        R[j]=s;L[s]=j;
    }
    td value=0;for (int i = 0; i < n; ++i)value+=cs[i][L[i]];
    return value;
  }
};
```

7.4 Dinic

```
// Min cut: nodes with dist>=0 vs nodes with dist<0
// Matching MVC: left nodes with dist<0 + right nodes with dist>0
struct Dinic{
  int nodes,src,dst;
  vector<int> dist,q,work;
  struct edge {int to,rev;ll f,cap;};
  vector<vector<edge>> g;
  Dinic(int x):nodes(x),g(x),dist(x),q(x),work(x){}
  void add_edge(int s, int t, ll cap){
    g[s].push_back((edge){t,SZ(g[t]),0,cap});
    g[t].push_back((edge){s,SZ(g[s])-1,0,0});
  }
  bool dinic_bfs(){
    fill(ALL(dist),-1);dist[src]=0;
    int qt=0;q[qt++]=src;
    for(int qh=0;qh<qt;qh++){
      int u=q[qh];
      for (int i = 0; i < SZ(g[u]); ++i){
        edge &e=g[u][i];int v=g[u][i].to;
        if(dist[v]<0&&e.f<e.cap)dist[v]=dist[u]+1,q[qt++]=v;
      }
    }
    return dist[dst]>=0;
  }
  ll dinic_dfs(int u, ll f){
    if(u==dst)return f;
    for(int &i=work[u];i<SZ(g[u]);i++){
      edge &e=g[u][i];
      if(e.cap<=e.f)continue;
      int v=e.to;
      if(dist[v]==dist[u]+1){
        ll df=dinic_dfs(v,min(f,e.cap-e.f));
        if(df>0){e.f+=df;g[v][e.rev].f-=df;return df;}
      }
    }
    return 0;
  }
  ll max_flow(int _src, int _dst){
    src=_src;dst=_dst;
    ll result=0;
    while(dinic_bfs()){
      fill(ALL(work),0);
      while(ll delta=dinic_dfs(src,INF))result+=delta;
    }
```

```
    return result;
  }
};
```

7.5 Min cost max flow

```
typedef ll tf;
typedef ll tc;
const tf INFFLOW=1e9;
const tc INFCOST=1e9;
struct MCF{
  int n;
  vector<tc> prio, pot; vector<tf> curflow; vector<int> prevedge,
      prevnode;
  priority_queue<pair<tc, int>, vector<pair<tc, int>>, greater<pair<tc,
      int>>> q;
  struct edge{int to, rev; tf f, cap; tc cost;};
  vector<vector<edge>> g;
  MCF(int n):n(n),prio(n),curflow(n),prevedge(n),prevnode(n),pot(n),g(n)
      {}
  void add_edge(int s, int t, tf cap, tc cost) {
    g[s].push_back((edge){t,SZ(g[t]),0,cap,cost});
    g[t].push_back((edge){s,SZ(g[s])-1,0,0,-cost});
  }
  pair<tf,tc> get_flow(int s, int t) {
    tf flow=0; tc flowcost=0;
    while(1){
      q.push({0, s});
      fill(ALL(prio),INFCOST);
      prio[s]=0; curflow[s]=INFFLOW;
      while(!q.empty()) {
        auto cur=q.top();
        tc d=cur.first;
        int u=cur.second;
        q.pop();
        if(d!=prio[u]) continue;
        for(int i=0; i<SZ(g[u]); ++i) {
          edge &e=g[u][i];
          int v=e.to;
          if(e.cap<=e.f) continue;
          tc nprio=prio[u]+e.cost+pot[u]-pot[v];
          if(prio[v]>nprio) {
            prio[v]=nprio;
            q.push({nprio, v});
            prevnode[v]=u; prevedge[v]=i;
            curflow[v]=min(curflow[u], e.cap-e.f);
          }
        }
      }
      if(prio[t]==INFCOST) break;
      for (int i = 0; i < n; ++i) pot[i]+=prio[i];
      tf df=min(curflow[t], INFFLOW-flow);
      flow+=df;
      for(int v=t; v!=s; v=prevnode[v]) {
        edge &e=g[prevnode[v]][prevedge[v]];
        e.f+=df; g[v][e.rev].f-=df;
        flowcost+=df*e.cost;
      }
    }
    return {flow,flowcost};
  }
};
```

8 Other

8.1 Mo's algorithm

```
int n,sq,nq; // array size, sqrt(array size), #queries
struct qu{int l,r,id;};
qu qs[MAXN];
ll ans[MAXN]; // ans[i] = answer to ith query
bool qcomp(const qu &a, const qu &b){
    if(a.l/sq!=b.l/sq) return a.l<b.l;
    return (a.l/sq)&1?a.r<b.r:a.r>b.r;
}
void mos(){
    for (int i = 0; i < nq; ++i)qs[i].id=i;
    sq=sqrt(n)+.5;
    sort(qs,qs+nq,qcomp);
    int l=0,r=0;
    init();
    for (int i = 0; i < nq; ++i){
        qu q=qs[i];
        while(l>q.l)add(--l);
        while(r<q.r)add(r++);
        while(l<q.l)remove(l++);
```

```
        while(r>q.r)remove(--r);
        ans[q.id]=get_ans();
    }
}
```

8.2 Divide and conquer DP optimization

```
// O(knlogn). For 2D dps, when the position of optimal choice is non-
    decreasing as the second variable increases
int k,n,f[MAXN],f2[MAXN];
void doit(int s, int e, int s0, int e0, int i){
  // [s,e): range of calculation, [s0,e0): range of optimal choice
  if(s==e)return;
  int m=(s+e)/2,r=INF,rp;
  for (int j = s0; j < min(e0,m); ++j){
    int r0=something(i,j); // "something" usually depends on f
    if(r0<r)r=r0,rp=j; // position of optimal choice
  }
  f2[m]=r;
  doit(s,m,s0,rp+1,i);doit(m+1,e,rp,e0,i);
}
int doall(){
  init_base_cases();
  for (int i = 1; i < k; ++i)doit(1,n+1,0,n,i),memcpy(f,f2,sizeof(f));
  return f[n];
}
```

8.3 Dates

```
int dateToInt(int y, int m, int d){
  return 1461*(y+4800+(m-14)/12)/4+367*(m-2-(m-14)/12*12)/12-
    3*((y+4900+(m-14)/12)/100)/4+d-32075;
}
void intToDate(int jd, int& y, int& m, int& d){
  int x,n,i,j;x=jd+68569;
  n=4*x/146097;x-=(146097*n+3)/4;
  i=(4000*(x+1))/1461001;x-=1461*i/4-31;
  j=80*x/2447;d=x-2447*j/80;
  x=j/11;m=j+2-12*x;y=100*(n-49)+i+x;
}
int DayOfWeek(int d, int m, int y){ //starting on Sunday
  static int ttt[]={0, 3, 2, 5, 0, 3, 5, 1, 4, 6, 2, 4};
  y-=m<3;
  return (y+y/4-y/100+y/400+ttt[m-1]+d)%7;
}
```

8.4 C++ stuff

```
// double inf
const double DINF=numeric_limits<double>::infinity();
// Custom comparator for set/map
struct comp {
  bool operator()(const double& a, const double& b) const {
    return a+EPS<b;}
};
set<double,comp> w; // or map<double,int,comp>
// Iterate over non empty subsets of bitmask
for(int s=m;s;s=(s-1)&m) // Decreasing order
for (int s=0;s=s-m&m;)   // Increasing order
// Return the numbers the numbers of 1-bit in x
int __builtin_popcount (unsigned int x)
// Returns the number of trailing 0-bits in x. x=0 is undefined.
int __builtin_ctz (unsigned int x)
// Returns the number of leading 0-bits in x. x=0 is undefined.
int __builtin_clz (unsigned int x)
// x of type long long just add 'll' at the end of the function.
int __builtin_popcountll (unsigned long long x)
// Get the value of the least significant bit that is one.
v=(x&(-x))
```

8.5 Interactive problem tester template

```
#Easier method with bash commands:
#mkfifo fifo
#(./solution < fifo) | (./interactor > fifo)

# tester for cf 101021A (guess a number, queries: is it >=k?)
import random
import subprocess as sp
seed = random.randint(0, sys.maxint);random.seed(seed)
n=random.randint(1,1000000)
try:
  p=sp.Popen(['./a.out'],stdin=sp.PIPE,stdout=sp.PIPE)
  it=0
  s=p.stdout.readline()
  while it<25 and s and s[0]!='!':
    k=int(s)
    assert k>=1 and k<=1000000
```

```
    if n>=k: p.stdin.write('>=\n')
    else: p.stdin.write('<\n')
    s=p.stdout.readline()
    it+=1
  assert s and s[0]=='!'
  k=int(s.split()[1])
  assert k==n
except:
  print 'failed␣with␣seed␣%s' % seed
  raise
```

8.6 Max number of divisors up to 10^n

```
(0,1) (1,4) (2,12) (3,32) (4,64) (5,128) (6,240) (7,448) (8,768)
    (9,1344) (10,2304) (11,4032) (12,6720) (13,10752) (14,17280)
    (15,26880) (16,41472) (17,64512) (18,103680)
```

8.7 Template

```
#include <bits/stdc++.h>
#ifdef BRAULIO
#define deb(...) fprintf(stderr,__VA_ARGS__)
#define deb1(x) cerr << #x << " = " << x << endl
#else
#define deb(...) 0
#define deb1(x) 0
#endif

// --- DEBUGGING TOOLS ---
void void_debug() { std::cerr << std::endl; }
template<typename Head, typename... Tail>
void void_debug(const Head& H, const Tail&... T) { std::cerr << "␣" << H
    ; void_debug(T...); }

#ifdef LOCAL
#define debug(...) std::cerr << "[" << #__VA_ARGS__ << "]:", void_debug(
    __VA_ARGS__)
#else
#define debug(...) 42
#endif

/*
Uso en competencia:
Compila localmente con el flag -DLOCAL para activar las salidas por cerr
    , por
ejemplo:
    g++ -std=c++17 -O2 -Wall -Wextra -DLOCAL main.cpp -o main

Dentro del codigo puedes escribir:
    debug(variable_x, mi_vector[0], "llegue aqui");

Cuando LOCAL no esta definido, como ocurre normalmente al enviar al juez
     en
linea, debug(...) se expande a una expresion inocua y no imprime nada.
    Esto
mantiene las trazas fuera del output oficial sin tener que borrar
    llamadas de
debug antes de enviar gracias a #ifdef LOCAL. Si tu entorno usa
#ifndef ONLINE_JUDGE, puedes mantener la misma idea: compila localmente
    con
-DLOCAL y envia sin ese flag.
*/

#define SZ(x) ((int)x.size())
using namespace std;
typedef long long ll;

int main(){
  return 0;
}
```